Welcome to this session on technical writing. In this video, we're going to explore what technical writing is, why it's essential in the field of computer science, and how you can develop strong technical writing skills that will help you to succeed both in your studies and in your future career. So what is technical writing? In simple terms, technical writing involves communicating complex information clearly and effectively, usually to a specific audience. It often includes instructions, manuals, reports, and documentation related to technical subjects like software development, engineering, and scientific research. Unlike other forms of writing, technical writing is focused on clarity, accuracy, and above all being concise. It's not about crafting a compelling story or showcasing your personal style. It's about making sure your audience understands the technical information you're presenting, no matter how complex it might be. Now you might be wondering why technical writing is so important, especially in computer science. Here are a few key reasons. Technical writing is essential for creating clear and detailed documentation. Whether you're writing software documentation, APIs, user manuals, or system architecture descriptions, the goal is to make complex systems and processes easy to understand for the end user or stakeholder. In the tech industry, collaboration is key. Teams of developers, engineers, and non-technical stakeholders often need to work together. Technical writing helps bridge the gap between these groups by providing clear communication that everyone can follow. It ensures that everyone is on the same page, whether they're writing code or making business decisions. Imagine you have to communicate an idea to a senior backend developer. Now imagine you have to communicate the same information to someone who works as a communications manager or in the marketing team. You have to utilize strategies that make your content as accessible as possible to as wide an audience as possible. Well-written documentation saves time. Whether you're onboarding new team members or troubleshooting issues, or implementing systems, clear documentation prevents misunderstandings, reduces the need for repeated explanations, and helps avoid costly mistakes. You will see this in other modules from the perspective of test-driven development or in sifting through design guidelines, but the key is to recognize diversity and improve accessibility of your content. Strong technical writing skills can set you apart in the job market. Many employers value candidates who can not only solve technical problems, but also communicate their solutions effectively. Whether you're creating technical reports, presenting project updates, or writing code comments, these skills are essential. There are lots of jobs in this space, particularly in the domains of publishing and user experience design. What makes good technical writing? Let's break down a few essential characteristics. The most important trait of technical writing is clarity. Your audience needs to understand the message you're delivering. To achieve this, use simple language whenever possible. Avoid jargon unless you're sure your audience is familiar with the terminologies. If you have to use technical terminology, be sure to define it clearly. Simple things like a glossary of terms can help make the content much more accessible. As a general rule, you should at least define an abbreviation the first time you use it. Be concise. Your goal is to communicate information as efficiently as possible. Avoid unnecessary details or overly complex sentences that could confuse your readers. Get straight to the point while ensuring that nothing essential is left out. You'll notice when we look at technical documents that most sentences are simple, with a handful of complex or compound sentences to break up the monotony. Technical writing must be accurate. A single mistake can lead to misunderstandings, wasted time, and even serious errors in a project. Always double-check your facts, figures, and instructions to make sure they

are correct. Imagine if I told you that this module was 1,500 notional hours of study instead of 150. I'm sure you'd have a lot to say about it. Good technical writing is well-organized. Use clear headings, subheadings, bullet points, and numbered lists to break down your information. This makes your writing easier to follow and helps your audience find the information that they need quickly. As a general rule of thumb, I like to go about four subheadings deep before it gets confusing. An easy way to remember this is to think about it like an IP address. 1.1.1.1.1 is invalid. Technical writing should be objective and neutral. You are presenting facts and information, not opinions. Avoid subjective language and stay focused on delivering accurate and useful content to your audience. You might get a little bit of opportunity to make some more reaching points in your discussion and evaluation chapters, but again you should be able to back this up with rigorous empirical evidence. I prefer dogs to cats is fine. Dogs are better than cats, while objectively true might need substantiating. Let's take a look at some of the common types of technical writing you may encounter in computer science. User manuals and guides. These documents are written for end-users to help them to understand how to use a product, whether it's software or hardware. The goal is to explain functionality in a clear and easy-to-understand way. Have a look at the man pages in Unix, or documentation for your favourite programming language, to get a sense of the flavour. API documentation explains how developers can interact with a software application or service. It provides details on how to make requests, what responses to expect, and any necessary configurations or dependencies. Most of what you work with will either be a framework or an API, and you probably already have lots of experience of this one. Technical reports are used to present findings, recommendations or project updates. This might include software performance evaluations, system architecture proposals or project milestones. These tend to talk to very specific audiences and are usually targeted as such. Presenting comments in your code is also a form of technical writing. Good comments explain what your code is doing, why certain decisions were made, and how other developers can modify or use the code. This can be difficult to master, but you generally expect to see short comments for relatively complex tasks. Where an algorithm or piece of code is a bit more complex, then you may see a more detailed description of exactly what is happening to help with understanding, maintainability, and even refactoring. Plug strings in Python are a really good example of this, explaining exactly what a given piece of code is doing. White papers and research papers. These are often more formal and in-depth, typically written for a highly technical audience. They present new research, findings, or approaches to solving specific technical problems. I've written papers on adaptive audio systems, digital gifting, hackable digital music instruments, and digital audio workstation usability. Each one of these had a slightly different style, tone, and level of technical language. What is common in these is that they facilitate an interesting discourse around some novel design, idea, concept, or technical proposition. I usually start by reading the abstract, then the conclusion. If both are interesting, I tend to read the rest of the paper. Make sure that these two sections really explain exactly what is going on in as few words as possible. Now that we know what technical writing involves, let's break down the process of writing a technical document into a few key steps. Before you start writing, consider who your audience is. Are they experts in the field, or are they non-technical stakeholders? Tailoring your writing to your audience is critical. If you're writing for developers, you can use technical jargon and dive deep into code explanations. But if you're writing for managers or clients, you'll need to simplify the content and focus on higher-level concepts. A

really good research paper can speak to lots of audience as well, but you cannot always get away from hardcore mathematical constructs or some inherent complexity. It's about communicating this in the most accessible way possible. It's better to speak to a million people than one person. What is the goal of your document? Are you providing instructions, explaining a concept, or summarising research? Knowing the purpose of your document will help you to stay focused and ensure that all the relevant information is included. Lots of research will start with a literature review, which brings in existing ideas. It is often later in the paper when you start to realise what the novel contribution is – something new appearing on the foundation of lots of existing, pre-established ideas and well-tested theories. Once you have your purpose and audience in mind, organise your information logically. Use headings, subheadings, bullet points and diagrams to structure your document. Start with the most important information and work your way down to the details. I find sticky notes are useful for this. You can then group them into themes and concepts to start to fold down mutual concepts into simpler constructs. Now it's time to write. Remember to use clear, simple language and maintain an objective tone. Start your introduction first, giving your reader an overview of what the document will cover. Then, move into the body, where you present your information in a structured way. You can start with bullet points or tables and then extrapolate from there. Once you've written your draft, review it carefully. Check for clarity, conciseness and accuracy. Edit any sections that seem too complex or unclear and correct any errors. It's also a good idea to get feedback from others, especially if you're writing for a technical audience. The peer review process helps with this when producing technical manuscripts and also acts as a check and measure for quality. There are many tools and resources available to help you improve your technical writing. Here are a few that you might find helpful. Markdown or LaTeX for formatting technical documents. These tools are widely used for creating clean, structured documents, especially for documentation and reports. We actually use LaTeX for formatting your examination scripts. You're also familiar with Markup from your HTML classes, which are also about building semantic structure into a page. AI tools for checking grammar, style and readability. There are even libraries in Python for natural language processing that can help with improving style or rewriting components. Graphics design software for creating visual aids like flowcharts, diagrams and process maps. API documentation tools for generating and documenting APIs. In this module, you'll get a chance to look at some of these, but as part of your working life, you'll be exposed to many more. To wrap up, let's review the key takeaways from today's video. Technical writing is essential for clear communication in the tech industry, especially when working on complex projects or explaining systems to diverse audiences. Some concepts cannot be explained in code or through mathematical concepts alone. Good technical writing should be clear, concise, accurate, well-organized and written with an objective tone. There are many types of technical writing, including user manuals, API documentation, technical reports and research papers. Always start by understanding your audience and purpose, then structure your writing logically and revise carefully for clarity. Developing strong technical writing skills will not only help you succeed in this course, but will also be an invaluable asset throughout your career. Thank you for watching and in the next lesson, we'll dive deeper into exploring the anatomy of a technical document.

Hello and welcome back. In this video, we're going to explore an essential document that will play a key role throughout your studies, your program regulations. Now I know this may not sound like the most exciting part of your learning journey, but trust me, understanding these regulations is crucial to your success. Our task today isn't to dive into every single rule and guideline in the document. It's our job as module leaders and program conveners to make sense of what can and cannot be done. But rather, this is to understand how we can break it down into something useful and accessible for new students like yourselves. It also means you get a good understanding about how things are structured and why we can or cannot do certain things. So what exactly are program regulations? In simple terms, program regulations are the set of rules, requirements and guidelines that outline how your degree is structured, what's expected of you and what you need to do to successfully complete your course. These include key details like the modules you're required to take contingent upon your pathway, credit requirements for passing and progressing through your degree, assessment criteria, for example, an exam, a coursework or a mix of both, the conditions under which you may need to repeat or withdraw from a module or a course, for example, if you get a grade of 20%. While these regulations are technical, they aren't tied to specific technologies or systems like coding languages. Instead they focus on the rules of the game that guide your academic journey. These often change, but you will generally follow the guidelines you've signed up for when you first enrolled. I've worked at seven different universities now in the general field of computer science, and you'd be surprised how much these can vary. You might be wondering, why should I care about program regulations? Here's why they matter. First of all, they provide clarity. Program regulations give you a clear roadmap for your degree. They tell you what you can and can't do, and they ensure you stay on track towards your goals. Understanding them early will help you avoid unexpected issues like not meeting a prerequisite or missing a key assessment. A degree is a pretty complex and time-consuming undertaking, so you need to know exactly what you can and can't do to get the most out of your time here. They help you to navigate challenges. Throughout your studies, you may encounter situations where you need to make important decisions, whether it's about changing modules, understanding reassessment policies, or knowing when and how to apply for special circumstances. The program regulations will guide you in these moments. The regulations set expectations. These regulations define your responsibilities as a student. They outline what the institution expects from you in terms of attendance, submission deadlines, and academic integrity. Following these rules is key to avoiding penalties or delays in your progress. Although you're working remotely for the most part, some on-campus degrees require a certain level of attendance or engagement. Perhaps we should make it compulsory for you to spend 100 hours on the platform for each module. No, of course not. That wouldn't be fair. The expectations have to be reasonable on both yourselves and us as the academic leadership team. Finally, they support your

success. Program regulations are designed to help you to succeed. They set the framework that ensures fairness and academic standards across all students. By understanding them, you're equipping yourselves to make informed decisions throughout your degree. These rules are for us as much as they are for you. Now for the practical part. Your task is to take the detailed program regulations for your specific course of study and summarise them into a short guide for new students. But why summarise them? New students, just like you once were, often feel overwhelmed by the amount of information they're given at the start of their course. A concise, easy-to-read summary of the program regulations can help them quickly understand what's important and avoid common mistakes. And summarising them is a great exercise in technical writing, a skill we've already touched on in the previous lesson. Here's how you can approach this exercise. Start by skimming the document. First, take a broad look at the full document. Don't worry about absorbing all the details right away, just get familiar with the structure. Look for the key sections like module requirements, assessment policies, progression rules and any special circumstances or appeals processes. What stands out to you as important? Is anything superfluous or just damn right useless? Identify the most important information. Your goal is not to recreate the entire document, but to pull out the most essential points. Focus on what a new student really needs to know to get started. What are the core rules? What should they pay attention to in the first year? Some key areas to consider are core modules and optional modules. What must they take and when? Assessment criteria. How will they be graded? What are the pass marks? Progression rules. What do they need to do to move from one year to the next? Reassessment or retake policies. What happens if they don't pass? Use clear, simple language. When writing your summary, keep it concise and easy to understand. Remember, this is an exercise in technical writing. Use bullet points or headings to break up information and avoid jargon where possible. The aim is to make it as accessible as possible to new students, some of whom might be unfamiliar with academic terms. Keep it objective. You're writing a guide based on factual information. Avoid giving opinions or advice. Stick to the key rules and guidelines that are outlined in the document. Be careful with text summarizers. I recently tried to get an answer to a program regulations question utilizing a fairly popular large language model, not saying any names. In the first instance, it produced incorrect information. When I queried this, it doubled down and lied to me. I didn't know the answer, but I knew that the answer it produced wasn't feasible. These tools are great, don't get me wrong, but please don't put too much trust in them. Let's talk briefly about the format of your summary. Aim to keep your guide short and to the point. Two pages should be sufficient. Use headings and subheadings to break the information into sections, such as modules and credits, assessment and grading, and special circumstances. Where possible, use bullet points or numbered lists to present information clearly and concisely. Can you turn everything into a single table? Think of this as a quick reference guide. It should be something a

new student can read through in a few minutes and feel confident about the key rules. This could even make up a part of your portfolio. It's a useful exercise and one you should have a go at. Now you might be wondering how this task fits into your broader learning in this module. Here's why this exercise is valuable. Summarizing a technical document like program regulations is a great way to practice communicating complex information clearly and concisely. It's an essential skill you'll use in many areas of computer science, from writing project reports to creating user documentation. Sometimes you might only have a page or two minutes to articulate some complex concept. Use that space or time effectively, and people will appreciate it. By summarizing the program regulations, you're actually reinforcing your own understanding of the document. You'll become more familiar with the rules that will guide your academic journey, which will help you to avoid pitfalls later on. Summarizing information and presenting it in a clear, structured format is a key skill when working in teams. Whether you're presenting findings to a technical team or explaining a process to non-technical stakeholders, the ability to distill complex information into clear summaries is highly valued in the workplace. To wrap things up, this task is all about helping you to develop the technical writing skills you'll need throughout your studies and future career. By summarizing the program regulations, you'll gain a deeper understanding of how to break down a complex document into something more accessible, something that will benefit both you and new students coming into the program. Take your time, focus on clarity, and remember to stick to the most important points. Once you've completed your summary, you'll be one step closer to mastering the art of technical writing, and you'll have a handy reference for the rest of your academic journey. Thank you for watching, and good luck with your summary. I look forward to seeing what you produce.

So, in today's exercise, we're going to look at some example documentation. So, we're going to have a look at my thesis document, which was conducted as part of my PhD. Now, this document is a fairly comprehensive one. It's an in-depth study, of course, it's a PhD, so it takes about three years to finish a PhD. It's quite a large body of work. The fundamental work of this was made up of three independent studies all linked together through some kind of thematic concept. And that's how a PhD is typically formed. It's three independent studies that are usually published, and then there's a narrative, a rhetoric that runs between those things that connects it all together as a thesis, as a document, which you can read by itself holistically from start to finish as a set of constructs around an idea, a research proposition that's then brought to fruition with three kind of independent studies that sit alongside this narrative that runs through it. So, those independent pieces of work are research contributions, usually novel contributions in that they are moving research on in some capacity or bounds. And then within that, there's a discussion, there's a discursive element to this, where you talk

about what the ramifications of these things are, what the impact of the research is, and you connect all the dots. So, you do the prerequisite work, then you go off and do some independent work, and then you step back and you reflect upon all of those things. So, that's what this document does essentially, is it summarises all these key points in a very specific and both general way. So, we'll look at both the specifics and the general aspects through this exercise to get a sense of how we could structure our own document and how we might want to engage in our own research to get the most out of the time that we have. So, typically, most documents will start with a title, a summary of what they're about. You'll have an author, some of the bits and pieces in there relevant to the institution, etc., etc. Now, the first thing you typically come across is what's known as an abstract. Now, we've probably heard this word before, we've probably heard this word used in a lot of different ways, but an abstract in the context of a document is essentially a way to summarise the key elements, the key takeaways from the document itself. So, you'll say, this is a research project about a thing, we explored these things, we did X, Y, and Z, and these were the general outcomes. Now, your abstract tends not to have things like citations in there, it tends not to go into too much detail, although there may be some specifics in there. So, for smaller research papers, for instance, you might include a summary of the number of participants to think about generalisability, or you might even include some general summary statistics in there just to give a sense of the kind of overarching themes and ideas conceptually and how they're being teased out as part of this process. The point of the abstract really is to help readers understand what the research is about without actually having read the research. And if you're engaging with a research activity, this is really useful for you, the abstract should be the first thing that you read. For me, when I'm looking at research papers, I'm trying to find relevant research, the first thing I tend to do is I'll have a look at the abstract and I'll say, okay, what is it that we've got in here that's useful to me? Am I actually going to go any further with this discovery process? Or is this outside of the scope of my research? So I might read this and say, yes, I think it's relevant. I'm going to continue trying to find the things that I'm looking for within this document, because I think there might be something here, or no, I'm going to discard this because it's not relevant to the type of thing I'm doing, or it doesn't cover the right methods or approaches. So when you're looking at the abstract, you're often trying to identify at a very early stage whether or not this is a relevant piece of research to what you're doing. So you typically be skimming for keywords, like, you know, you might find software packages and go, okay, I think that's relevant. You know, they engaged in some methodologies that implored things like workshops, discussion groups. Okay. They were looking at aspects like user satisfaction. So straight away, you can get the general themes and ideas from the research. They've already started to impose some structure on the research here as well. So I can see that there's some analytic lens, some discursive lens. They've imposed some heuristics or created some heuristics from

scratch. So again, that will help me to understand if this is relevant. Now we're not going to go through this in order. We're actually going to go through this as if we are external researchers. So I'm imagining I'm reading my thesis as somebody who's coming along, maybe starting my own PhD, or I'm an undergraduate student who's trying to identify some key research. So the next thing I do once I'd read the abstract will be to go ahead and I go and read the evaluations and the conclusions. So I'd skim all the way down to the bottom and go, okay, what are they talking about right at the bottom of the document? And if I go down here, I can start to see some of the strengths and limitations of the work. I can identify some of the key outcomes, the summary findings. And then from there, I might decide that I'm actually going to spend some time looking at the research in specific aspects of what they've done and take their approach. But I would start typically with the abstract, then move on to the evaluation. Then I might have a quick skim through the introduction. And then finally, I might read the whole paper. So that's a good way to think about your research in starting with the abstract, moving to the end of the paper, looking at the evaluation, and then skimming through to see if there's any other key aspects before you go ahead and commit a huge amount of time to researching and reading in this space. So that's a good way to approach your research. Now within the context of this document, we're really interested in the structure too. So we're going to spend some time talking about the structure. So as we scroll down, we can start to see that a structure has actually been imposed on this document. So this is really helpful when you're producing your own documents to help other people navigate the information architecture. It's a bit like when you're building a website, you have that navigational structure along the top or down the left hand side, from top to bottom or left to right, which usually gives you a sense of what the key themes and ideas are. So if you're looking for groceries and you're on a grocery shopping website, you'll see some general headings, the sections that they think will be relevant. So you might have a section for dairy, or you might have a section for kind of staples, key items that you purchase every week, and so on and so forth. Same with the document, it has an information architecture, and it should have a structure that looks roughly like this. You'll notice within this, we've got some tables inside our tables. So we have a table of figures, which is a key one. And we also have a table of tables, which sounds like a really meta way of doing things, but actually being able to navigate the key aspects of the document, particularly if you want to make reference to these later on, if you're doing external research, that's looking back at this stuff, having access to these things. Because for me, these are some of the key outcomes of the research. Quite a lot of the time, the key outcomes from research are either represented visually as figures, or as that kind of XY structure. How does X compare to Y? How can we compare these two things to one another, and start to identify at that analytical level, key things that are going on. Within this structure, we've also got other things. We've got chapters, and then we've got headings and subheadings. So we've got a numerical

format for this. So chapter one is of course, the introduction. That's where we'll have lots of research, we'll have our literature review, we'll have some formal outcomes that describe what's gone on in previous research, that connects the contemporary stuff that we're doing here, to the stuff that's happened previously. So we're looking back and going, these are all the other bits of research that have happened in this space. This is where we're going to move from. And then you start to move towards the more formal aspects. So things like your methodology, your approach to the research, and then the actual research itself as you move through the chapters. But the early phases are really about identifying those key aspects. So what you might see with a document is a formal introduction chapter, you might see a formal literature review chapter as the first thing. It might be interspersed within the document, so there might not be a formal literature review, you might find that it falls across lots of these early categories. But generally speaking, for most research documents, you'll have a literature review chapter, particularly for long ones. For shorter papers, the literature review might be included in the introduction chapter, or it might actually stretch across multiple chapters. If for instance, it's a mixed methods approach to research, where they're trying to tease out lots of complexities, and the discussion is emerging from a connection of the contemporary, the contemporary research, and trying to connect those dots together to understand what the broader implications are. So again, this is a stylistic consideration. It's not that you have to do it one way or another, it will largely depend on what field you're working in. But when you're writing your early research papers, I'd probably suggest you start by writing a formal literature review chapter, probably quite early on in the process, to help to set the scene for the work to come. If you create these citations early on in the work, it allows you to refer back to them as you go, which can be really helpful, and it frames your discussion around what's already happening. Now scientific discourse isn't about necessarily creating the newest, most interesting thing, but you're actually drawing on lots of resources to try to bring them to the point of discussion at which you can then start to tease out some new, complex bits of information. But you have to look at that prerequisite first. Before trying to identify new things, you need to look back at what's already been done to substantiate your claims, to have a look to make sure you're not treading on somebody else's toes, you're not copying work that's already been done, that actually the work you're producing is new, is interesting, is novel, is contemporary, all of those things. And to do that, you have to frame that against an existing structure. And in the context of most research exercises, this is a chapter that includes lots of citations, i.e. a literature review or some formal chapter to describe those processes. Now within each of these sections, we've got subheadings. So for the shorter sections, for instance, we have just a handful of subsections here. So we've got subsections one to six, and we've only gone two subheadings deep here because it's not a particularly introspective discussion. There's not a huge amount happening here. So there's two subheadings are enough to allow us to dip down into a

slightly lower level of discussion. So we can talk about the structure, how this document is structured. We can talk about the chapters that have some kind of investigative nature, some aims, objectives, etc., etc. So we start at a fairly high level here in the first chapter. We only go down two subheadings deep because the discussion is not particularly nuanced. When we move to chapter two, we start to see some content appearing so we can see the gaps between the pages getting much longer as we're starting to kind of build up speed. We're starting to accelerate more quickly and move on to new concepts and ideas more quickly. Because this is a literature review chapter, that's not particularly surprising. So we talk about things like the early usability research from the 90s, characters like Jakob Nielsen, Donald Norman, Ben Schneiderman, the kind of fundamental research that underpins this work. And quite a lot of the time, you'll find that the early work in your literature review goes back as far as you possibly can to the very early pioneers in the research space. And then you move towards the more contemporary things as you kind of step up the discussion. You'll also typically have a methods chapter in a lot of research articles. Now, you might not necessarily see these in every type of article that you engage with, but generally speaking, you'd have some kind of methods or methodology chapter that would describe a process. So this is typically like a set of ingredients could be compiled in such a way as to give instructions on how to make those ingredients into a meal, into a particular substantial version of those smaller component parts. So the methodological approach is really just a set of instructions of how to turn those ingredients into something substantial, i.e. a meal, or it might be how to bake a cake or something along those principles. And the research methods chapter is essentially that it's a way to encapsulate the approach that you've taken to research. Traditionally, this was thought of as a way, a means by which the research could be replicated. If you imagine scientific experiments, that's the way I tend to conceptualize this. So your science classroom, when you were studying science as a student, you may have found that you had lots of beakers and you were kind of heating things up and combining chemicals together in different proportions. Well, methods chapter is essentially a description of that process. So what you did at each step at each phase, how you combine things in what proportions, the methods chapter tends to be heavily driven around the empirical data that you're collecting. And largely for computer science-based studies, what you'll find is that there's lots of metrics imposed on things. So ways to measure things. So we measure things according to some structure, centimetres for height, for instance, we might measure weight in pounds or kilos. So we have those approaches that we can take to trying to encapsulate the approach that we took to research so that other people can think about the environment that you engaged in, the apparatus, the resources, the process, the step-by-step process. For instance, for my research, I've got a category for open coding where we identified core themes and we started to try to fold those down into smaller conceptual ideas. So a series of themes of the research, we look through everything that we do and

then we collapse things into structures. So we go, okay, there's five core themes in here or there's 10 core themes in here, 100 core themes. And then we start to fold that down even more into something that vaguely resembles a structure or an ontology. So within the structure, again, we tend to focus on subheadings that are appropriate to the level of depth of the discussion. So here, because there's lots of new themes emerging and these are all categories by themselves, we actually go another subheading level deep. So we go from 3.1 up here and 2.5 that we're using here to three subheadings deep. So 3.1.1 as it becomes more detailed, as it becomes more granular in terms of focus. And then the research itself kind of summarizes each section. So this is something you'll probably want to do as part of your own research. When you're giving your document structure to start out with things like considerations and end with some summary within those sections. So you don't just have a summary section right at the end of the document that tries to bring everything together. You have a summary which kind of leads into the next section to give the document some kind of emergent flow. Say, OK, this is what we found out. This is what we're probably going to do next. And then the introduction chapter for the next chapter can actually build upon that. So we go, this is where we're going. OK, we're going to start on this new journey now in the introduction. And that's the idea behind this process. So as we move to chapter four, we can start to see some empirical work being done. We have those three subheadings deep and we look at the more detailed focus thing again. It's starting to become a rather lengthy document now. So we have to impose a new set of structural components on this. And for the empirical work that we've done, we've got categories and categories inside categories and categories inside categories inside categories. So we can see that there's a structure imposed upon that there. For each one of these, they're kind of separate studies. So what I was talking about there, the idea of three independent studies, you can think of each chapter as a study. Each chapter is essentially a paper. So this could be a research paper. Chapter five, ethnographic exploration of music systems in context could be a research paper. And in fact, quite a lot of the work from this was published in some way, shape or form, whether it was a poster presentation of some of the ideas or concepts at a low level to the more kind of empirically rigorous. data at a high level, at maybe a major journal or conference proceedings. So again, a PhD thesis or something substantial, maybe a book that you're reading, might be made up of lots of smaller components, smaller papers, smaller bits of research that are all brought together to have a more substantial engagement with the discourse. And this is essentially what a PhD or a book or something of that level of depth is really covering. Whereas you might also see short papers or works in progress, which could be one or two pages long, for instance, which is one key thing that's just been discovered or is emerging. And both are valid types of research. The level of discourse usually depends on how contemporary it is and what's happening within that space. So a shorter paper might be a new discovery in AI or machine learning, for instance. A longer paper might

be a more kind of interesting systematic discourse around the tools and techniques used in the space. So again, each method is appropriate for purpose. Each mode of research, as we see them, has some general way of approaching the research space and also a way to communicate with a larger audience. So again, this is a structure we see, so lots of larger chapters now with sections and subsections. The key thing I want you to take away from this is that when you're imposing some structure on your documents, or when you're trying to interpret a document to understand its structure and get at the stuff that's useful, when you're trying to skim through this and find things that are relevant, what you want to do is take a systematic approach to it. So as I said, start out with the abstract, which gives you a general summary of the research, then move to the evaluation chapter. So you start out with the abstract, move down to the evaluation and read those things. So the evaluation and conclusion would be the second and third thing I'd probably read if I was looking at this externally. I'd get to the end of that and I'd have a general sense of not just what the research was about, but the approaches they've taken, the methodology, the outcomes, the interesting discourse. From there I could make a decision about whether or not this is worth referencing, whether or not this is worth reading, spending my time trying to discover more about this research, or whether it's irrelevant to my discussion. So again, what you're wanting to do is try to maximise your time and be efficient, make sure that when you're looking for these documents and you're trying to find things in your own literature reviews and your searches, that you're able to discern the research that's valid for your specific research question, versus things that are irrelevant. And of course, when you're creating these structures yourself for other people to engage in, to be a good scientist you want your research to be as accessible as possible. So pointing to these sections where people can find things, even with a large document like this which has quite a bit of substance to it, you can still impose a structure to this so that people can navigate it and find the interesting bits of work. So if somebody's only really interested in how you've approached the problem space of creativity in requirements elicitation for collaborative work, for instance, which is a bit of a mouthful, you can scroll down to the section on that which is chapter 6 to say, okay, they've done some requirements elicitation work here, they're trying to understand what people want from these systems. You can then scroll down to the approaches, you can see how they've embedded different types of creative approaches to requirements gathering, and then you can navigate to those sections. Rather than just sitting down and reading the document as a whole and saying, okay, I've read 177 pages now and realised this isn't relevant to my research, I've probably wasted a day's worth of reading and I haven't found out anything useful. So again, you're going to want to spend lots of time trying to discern meaning from things as a good researcher, as a good scientist, and so you want to maximise the amount you can get done in that time. Instead of spending, you know, 40 hours a week trying to work on a problem space, this might take it down to 10 hours, 5 hours, or

maybe even 2 or 3 hours, and then you can spend the rest of your time more productively. So if you're a researcher who's writing papers, who's writing out research, who's formalising document structures, make sure that you embed a structure that people can access that's accessible, easy to understand, where people can elicit things from it quite quickly and navigate this structure in a way that's meaningful. In the same sense, if you are looking at this as a kind of external researcher, so you're wanting to engage with other people's work, make sure you're embedding these processes. Make sure that your research includes things like these nice figures to help to understand things at a broad level. Something like a figure or a table can help you to understand things like structure more easily. So here's a nice example of how I did it in my research. So I had three separate studies which described the process, and then within each of those studies there was a level of depth that was required. So for one of those axes or axioms, as you might describe them if you're a mathematician, there is a process of trying to create something that's a deliverable, a deliverable component from this research, and there's one that focuses more broadly on the contextual implications. And as you work through these axes of going from the kind of basic specialised focus of trying to understand the quantifiable data, so essentially just taking lots of numbers and trying to compare and contrast, that's a level of depth that's not particularly interesting. We're very focused, it's very specific. And as you move down this trajectory, as you move down towards the more broader implications, we actually spend some time with larger groups of people trying to elicit the broader set of requirements versus looking at those very, very specific use cases in the first instance. So again, you take a figure like this and it can describe a process nice and easy for you, it can make things easy for you to understand. And what you want to do if you're a researcher is to produce lots of figures and tables to help improve the accessibility of your writing. And in the same sense, if you're an external researcher looking at these bits of work, you're going to want to use these to form the basis of your research. You'll find you're often referencing figures and tables in a quite systematic way to encourage the discussion to move further. So science really is about bringing these things and taking them to the next level, saying this person discussed X, so that might be study one. This person discussed Y, which could be study two. And then finally we'll move on to something that's more contemporary. So we'll say based on X and based on Y, we can move forward and start to investigate Z. And what that gives us is a nice framing for our research. It provides a discursive atmosphere where we can validate most of our findings through external research and then create something new that helps us to move to that next level where we can say okay, now as a scientist I'm doing something that's original. I'm making an original contribution to science. And generally speaking, if you want to publish work, that's the type of criteria that will be required for you to get those publications into place. So in terms of your discovery process, in terms of your engaging with research and writing research papers, try to follow a structure that loosely

compiles in this manner, where you've got a nice descriptive abstract, where you've got lots of structure including tables, figures, nice headings to help people navigate with numerical representations of the pages that they can find things on, and then some structure imposed within that with sections and subsections where it's relevant. This will really help external researchers engage with your research and it will really help you in terms of actually structuring your research in a systematic methodological way, which is helpful to frame your own mindset and help you to think about what comes next. If you write these chapters up in formal sections, it will help you to understand what you've already covered and what you need to be moving on to next, based around your own discussion. So as I said, you know, you have an introductory chapter at the beginning of each section, you have some summary outcomes at the end, and it will help you to then settle that idea down, to put it down on paper and to move on to the next idea. So again, there's lots of reasons to do all these things that I've suggested. We're going to have some activities where we have a go at this ourselves, where we think about the structure of documents, but for now this is a nice primer to help you along the lines of trying to understand both how to write good research papers and how to engage with the wider discourse of research that's available to you as an academic.

So I'm going to try to describe how to take some really complex information and turn it into something meaningful. Now, the way that we often do this with research projects is we try to take really complex data and turn it into fairly simple visualizations that people can read and interpret and understand. Now, this is more difficult than it looks. It sounds like a simple process, but I'll talk about why this is such a difficult set of activities to engage in, and some of the tools and techniques that we use to do this. So I'll talk about some of my practical experience of translating a fairly complex set of processes into something that derives meaning from these processes. So the project I worked on to do a lot of this work was called the FAST project. It was the Fusing Semantic Audio Technologies project. It was about a £6 million project. It was a big research project done with EPSRC and in collaboration with a number of universities and industry partners. But the process behind this was trying to discern meaning from existing processes and look for design opportunities to create new or better systems. So we explored music making practices and looked at the production process to try to understand opportunities for improvement and opportunities for innovation. So there was a real focus on understanding the how of music making, how music was made, the methods that people engaged in, the practices that both musicians and producers engaged in to create successful musical products. And the fieldwork encapsulates the complex practice that is producing music. So we did a lot of fieldwork to understand those aspects of how and why. In many ways, the quantifiable data that we use from studies can be really, really helpful. But sometimes we actually have to spend some time in situ to really understand some of these subtle complexities. The things like how

people are motivated to do things, what motivates them to do a particular thing in a particular way, and so on and so forth. And so when we design and describe systems, we often do that from the perspective of having a really good understanding about what our users are doing, how they're engaging with systems, and what the potential for improving upon those processes is. So these complex field reports give us a starting point for trying to discern the practical implications for design. Now the challenge behind this is that even when you've done these complex field-based reports and you've produced these 300, 500 page documents, the raw data in this qualitative format still lacks meaning. So you can describe what people are doing and how they're doing it, but you can't necessarily discern a set of design goals from it. You can't necessarily propose a system from these complexities. Because the documents are so complex and there's so much embedded within them, you have to spend some time really understanding them and translating them into something that somebody else can use. So these types of documents typically describe actions and motivations, things people are doing and things that they'd like to achieve. As an example of this, we might see somebody clicking on track two and then they might adjust the microphone position. They might then say something like, that sounds better, the bass response is better in the mix. Now what this essentially means is that there's an improvement in this process. So they physically change the environment, they physically change the position of a piece of hardware, and that's translated into an improvement in the overall process, which is good. So we can see from this, from their actions and the thing that they've said about their actions, that there's some positive reason for doing this. So this is what we mean by design opportunities. There is an opportunity here, for instance, to automate moving the mic into specific positions. So maybe that's a design implication for something like a robot or something like some automation in changing the configuration of a system. It gives us some room for improvement. But this isn't quite useful for designing a system yet, because we don't really understand if this is something that's generalizable, if it's something that's common that happens, and if it's going to be useful for more than one individual user. So we want to see that this is generalizable in the sense of it having more than a single use case. So this is typically what this work looks like. So you might have somebody playing an instrument, it's plugged into a physical interface, they're listening to it as they play. They're also clicking around in the interface to change the sound of the composition. So it's not an easy thing to do, because what you're often doing in this process is what you're listening to on the instrument in the kind of raw format it's in. For an acoustic guitar like this, for instance, with some digital input, what this will be is you just jamming around, just listening to the chords and how they sound in your natural environment. When you apply things like microphones or effects or settings, how that sounds in a digital context changes because you're moving from the analog space into something digital. And so there's a lot of work that goes into clicking around this interface, what we refer to as the digital

audio workstation. You can see this in the bottom, where we've got that nice long waveform, which describes the audio. It describes the composition in its current format, and we apply lots of effects to things. So we will stretch this out and change the pitch and do lots of things to try to adjust this to make it sound better for a digital environment. Just remember, that's how people are going to be listening to this. This isn't something that's done in a live setting. As you can see here, I'm playing my guitar, but there's nobody else in the room. It's just me. I'm not playing just for my own purposes. I'm recording something, and so it's going to go somewhere. This could potentially be streamed by millions of people. So unless you know the audience, you're not really sure how to translate this into something digital. But this process of changing the effects, messing around with the composition in this digital format is really thinking about the end product of how this is going to out, how it's going to be listened to by a group of people. So there's a process of translation from these complex documents describing that example we saw in the last slide, where we're trying to produce something for purpose, and we're looking at three things effectively. We're looking at existing best practice, so things that work, things that are working well in the system that we want to copy effectively in our design of a new system and say, those things are good. We want to emulate those practices. What you might find is this comes largely from your market research. So if you're working on projects and doing these types of things, you might find you're looking at similar things that your competitors are doing. So a classic example of this is blue underlined hyperlinks. We generally see this as blue underlined. Wherever we go on the web, everybody follows the same conventions and practices. It makes sense to emulate those practices in most settings. We don't want to reinvent the wheel too much because there's no reason to do that. It would just create confusion. So we want to embed best practice in what we do. We want to look for opportunities for improvement. So if there's a problem with an existing system or something isn't as effective or efficient as it could be, then we look for ways in which we can improve upon those processes. And then finally, when looking through these research reports, we're also looking for opportunities for innovation, where we can create a new system that will help improve upon existing processes in a new and interesting way. What we often call the novelty proposition. There's something new coming from this process that allows us to create a new type of system, a new type of algorithm, a new type of interaction that will help to improve upon these processes. So the example might be producing a robot that changes that microphone position for us. That's an innovative solution to this problem. In many cases, you won't be working in innovative ways, you'll just be thinking about improving on existing processes. But there are always going to be a handful of design improvements that are possible through innovation alone. Now the real challenge with this type of stuff is that even when you do these field reports and you embed this complexity in these complex documents, they're still very hard to read and interpret things from. So you have two problems, really. The

first is you've got a high volume of information. With field-based studies, there's usually a lot of observational work that goes into them. They're very long, they're very cumbersome, they're not very accessible. It's not the easiest way to interpret and access information through these long reports where you just have to read about the very mundane details of day-to-day stuff. When you're engaging in these types of field studies, what you'll often find is that you have to report everything. You don't get to make the decision about, I'm not going to write about this because it might not be useful down the line. You capture everything and then you decide further down the line whether or not you're going to use these things. So you're looking at high volume of data and also high complexity. So they might be talking about technical things, there might be technical processes going on. You have to try to understand both of these things when you're producing these reports and then even more so when you're translating them into some other format. So the subtleties and the nuances are really where you want to spend most of your time. So when you look at these kind of complex reports and you read three or four of these really complex reports, if you have the time, then there's really only two ways that you can visualize this complex data. And there's two dimensions that you can describe this trajectory of data across. So the first is really about actions and intents. So if you're used to using system design activities in your life, you might find this manifest as a use case or some kind of activity diagram which describes a behavioral process. These are quite common in lots of agile processes in looking at use cases, often referred to as user stories, which describe largely actions and intents. And then more broadly, you might want to look at the state of the process. So something like a sequence diagram will be an example of this. But in terms of this process, you're looking at both actions and intents and state of the process as being equally important. So imagine you're driving somewhere, you will generally engage in a series of actions and intents. So that will be you mirroring, you signaling, you maneuvering, you engaging in all of these processes that are mechanical in nature. There are actions and there are things that you're hoping to achieve by those actions. So you press on the accelerator pedal with the hope or the intent that the car will move more quickly. And the additional pressure that you apply to that pedal will increase the speed you're traveling at. Now those mechanical processes, they're very important. You need to know the mechanical process of driving, but you also need to understand the state that you're in when you're driving a vehicle. So you need to know where you are, you need to know where you're going, you need to know how close you are to the target destination, and so on and so forth. So an example of this might be the car itself manifests lots of actions and intents that you can engage in for different driving conditions. So you might want to change the mode that you're driving in. If you're driving in a sports utility vehicle, for instance, you might want to change the mode if it's snowy to make sure that you're using all four wheels to give you that drive to make sure that your car doesn't veer off the road. You also want to understand the state of the process.

So you need to know where you're going, you need to know all the restrictions, you probably want to know how long it's going to take you as well. So if you're stuck in traffic, you're late to work, for instance, then you're going to know how late you are. So you can call up and say, I'm going to be 15, 20, 30 minutes late. So again, both of these types of information are really useful, but in different contexts of understanding what you're trying to achieve. Now the flow of information is something that looks like this. So we take these field reports and we translate them into steps or stages in a process. So they start out with initiation. Somebody has an idea for a song, they share materials together, they rehearse, they practice together, they locate a place that they can produce the music, they circulate parts of the composition. This is very boring and mundane, and this is a level of abstraction above the detailed reports that we'll see. But we need to understand the steps in the process. And we need to look at these series of reports to understand generalizable criteria about how people engage with the system in a flow oriented way to understand step one, step two, step four, step five, step 100. If we're designing systems for people, then we need to understand that flow of information. It's key to helping us to describe the types of systems that they might want to engage with. Now the concerns are also important. It's not just about actionable insights, it's about understanding the underlying conditions in which this system is going to work. So here's a quote from a participant from one of the studies. It says, verses for me are where the art form comes into it. I always try to be conscious about how layered my verses are with double meanings and conscious thought. The wording in verses for me is very important and I'm often caught in balance between keeping a rhyming pattern and saying something clever, meaningful, or just funny. Therefore, I build my verses line by line, often like a puzzle, where I know the structure it needs to keep to sound easier on the ears. I just need to find the right words to fit in the structure to express my point. This tells us a lot. Something simple like this, a simple quote that isn't particularly long, can tell us a lot about how to design a system. So we know that the participant has a preconceived notion about what they're going to be producing. They have an understanding about the architecture, the structure. They've got a general sense of certain words, the play on words mechanism. Their view of this, this analogy of a puzzle I really like, because it's about thinking about how the pieces fit together. A composition of audio is not just about it's one layer of things. It's one component part. It's often lots of instruments layered on top of one another, lots of vocal elements together, harmonics, etc, etc. So we take these concerns and preoccupations alongside the activities they're engaging in, and then we try to match them together in a meaningful way. So again, it's getting even more complex at this stage. Now the practical implications of this might be something like this. We've got a nice visual here of how the system looks to them. They've got a series of compositional component parts here. So we've got some drums, we've got a kick drum, which is a bass drum. We've got a snare, various other instruments in there. And the levels for each instrument are

set at a particular amplitude. What you'll often find with these compositions is they're also panned in a particular way. So one sound will be coming out of the left side speaker, one will be coming out of the right, one will be more central. So the configuration of the audio is done inside the interface. So there's some practical ways that you can do this. And we can see from the actual practical elements of how they interact with the system, what they're trying to retrieve through these compositional processes. It's not just about that thing that they described. It's not just about the things that they're doing, but all of this information together, which is really rich, contextually nuanced. Now the starting point for this, when I started visualizing these complexities, was really understanding the mapping of these two things together. So we looked at concerns and preoccupations, and we looked at practices, and we tried to find a way to map the two together. So the commonality of, I am concerned, or I have a preoccupation with doing this thing. And then finally, the practical ways in which they manifested this to come about, making things come to fruition through practical strategies. I click this, I did that, versus I want to achieve this, and I want to achieve that. And then we looked at generalizable properties of these things. So you can see that table at the top, which describes the generalizable properties of things like project initiation. They all contained a series of concerns and preoccupations coded with O. So you can see O3, O4, O5, O7. And then they map to a set of practices. In that example, we have P3, P16, P17, P19, and P29. So the idea behind this is that we were able to look at what they were trying to do, and why they were trying to do a particular thing, and then the activities that they're engaging. This gives us a really nice starting point for translating those really complex details into a meaningful set of visual things that we can do to describe these processes. And this is the first starting point for this. This is the first useful visual we can provide, which is a tabular structure, a nice X, Y representation, where we map two things against one another so that we can start to see patterns. And each one of these two tables has an X, Y representation. We map concerns and preoccupations against a series of practices, and then we map a series of stages of the project from project initiation right the way through distribution as both a series of occupations or preoccupations, concerns, and practices. So again, we're looking at this mapping between objects to help us to discern interesting bits of information as we go through. Now the next step we can look at this is generalizing through a series of processes. So trying to move up a level of abstraction from, okay, we looked at those very specific things that the people were doing. We look for common trends in their behaviors, and then we map those common trends across a series of activities that are generalizable. So we've got a concept of pre-production, production, post-production, and delivery phases. So these are the general processes that people engage in when they're making music. If we look at the table, we've got a nice visual structure just under the table that describes that process. We've used the color scheme for that, as well as a numerical scheme to help us in describing those processes. So again, we start with

project initiation, but that comes first, and then we finish up with distribution. And there are steps and stages that have to happen between those processes that we're then able to loosely group up and say those things have commonalities. So for instance, in pre-production, we start with step two, which is planning and preparation, and we finish with signing the sound. We then hand over to the next phase, which is production. And then at step eight, we hand over to post-production. And then in step 13, we move over to the delivery phases. So we've taken all those complex details, mapped them into a series of tables, taken that then and produced a clear visualization of steps and stages using both colors and numerical formats to describe. And then finally, we've got this really nice abstract representation at the bottom, which takes those thousands of pages of research reports and translates them into just five steps. Now, I think that's quite incredible that we're able to do that and discern meaning from things in such abstract ways. But that's not particularly useful in the format it's in either. So we need to think about how we can translate that into something more meaningful. So we can look at specific processes therein, or the processes that are contained therein, and try to discern meaning from each individual set of processes, while still keeping in mind that there are practical implications about the entire process happening from start to finish in a very specific order. So when we communicate this process, when we're trying to translate this to a set of design requirements, or perhaps to a group of users who are designing or developing this type of system, then we want to first of all simplify, but then develop a coding scheme and then group logically related items together. So we can look down at those detailed elements and visualize those, as we've done here, as a series of processes in collaboration, looking at essentially a conversation between two different artists of technical level, and looking at how they have that conversation and how they're able to collaborate in a social way to engage that process of collaborative music production, and look at that as a set of practical activities, and then look at how that fits against a broader scheme. So you can think of this like a standard. We've defined and described a standard protocol-oriented way of working. So we've got a series of activities that happen that are generalizable, that we can translate up to an abstract description of the processes and specifics around the orientation of those actions. So now we can step up to a level in which we're able to embed design interventions. We're able to build systems from this. So we can look down at each one of those parts of that complexity, that's still at a level of generalizability, where these aren't specifics about this one person does this thing, more these are the series of activities we tend to see within each of these phases of production. So for example, when we describe distribution, the distribution of music for purpose, we think about this as commercialization often. So people listening on streaming platforms, people going to buy a record or a CD, but there are inherent steps and stages within each one of these individual set of processes. So it depends on where this is distributed on how people engage with this type of system. So for instance, for the social media platforms, there's

a process of archival, there's tagging of metadata, which is very important, the information about how this is produced, where it's going to go, how it's going to be communicated in a kind of semantic way, versus just the composition. Actually, this is now a digital artifact, which has all this metadata attached to it about the original artist and where it's come from and where it's going. There's then a process of a manifestation of activities again, so uploading the audio to the platform, posting to various social media channels, reposting in various other places and then these feedback loops. And then we can drill down another level and we can say, well, okay, what happens at these feedback levels? Well, there's a process of engagement in a socially augmented way online to communicate about your own music, about your own composition, how people feel about it, how people are listening to it. So we can then look at this as a broad visual structure to help us to understand where all the design opportunities fit in this broader context of things without spending lots of time looking at the mechanical actions of, they click a button, then they move a slider, then they drag something across the screen. This is discernible to somebody who's a designer or a developer to say, actually, there's some actions I can perform to create a system that will facilitate some or all of these activities or improve upon existing activities that exist. Now when you take this and you talk about a community of activities, you can also look at this as a kind of set of research outputs and say, well, let's talk to a wider community of people about problems with existing systems. Let's talk about how people work with existing systems. And we can do that in this context. So one of the outputs of this research project was the description and definition of what was called a digital music object. Now the classical way we think about music is often we hear a track on the radio and we sit down with a friend or a colleague and we say, did you hear that track? Did you like it? What did you think about that other catalog of music? It creates really interesting conversations. Music tends to be something that's quite a socially augmented activity. Within this context of music production, the idea of a piece of music being a composition that sits alone by itself is probably a dated idea. And if you look at how music is produced and consumed, there's lots of activities that people tend to engage in at different levels. So there might be remixing, there might be reproduction of things. There might be a process by which something is fixed or improved upon. So you often see that tracks that have been created 30, 40 years beforehand are then reproduced again using contemporary technologies to create a better mix, to create a better version of that song. So there's all these things going on, again, sometimes happening synchronously, sometimes asynchronously, that can really help to embed this augmented social practice that I think improves upon music, generally speaking, if music is embedding the best practice of existing music, but also innovating in some new ways. It means that music can constantly be produced that's both new and old in the way it's defined. So we could think about this in a very, very simple way. Again, this is a visual way that we could describe this. We start out with the idea at a very basic level

that music is a composition, that we sit down and we record something on an instrument, and that's the end of the process. But actually, when we look at these field-based studies, we can discern a lot more from them. We can see themes, we can see people engaging in repeated activities, we can see processes which are happening in multiple spaces. So the composition could become some sheet music, it could become some notation, it could become a recording which then might change a number of times over its life cycle. So that recording could go out to various different places in various different formats, YouTube, for instance, SoundCloud, various other platforms. For YouTube, it might be a video format, which could then be remixed by somebody else, and then somebody else beyond that process could take that remix and create an acoustic version of it. Somebody else could then come along and remix the acoustic version of the remix of the video. So what you'll find is that the composition could change through its life cycle. And this idea of a digital music object came about through these observational studies of how people interact with these types of systems. So this is really interesting. I think it gives us some sense of actually music isn't this static thing that just exists. We just create it, it exists, and then over time it just disappears. These are often practical things that exist and then get changed and manipulated and recreated and chopped back up and then put back together in interesting ways. All these different processes that help us to understand that music is evolving over time. So we started out with the idea of all these complexities around things like processes. So we see these on the left hand side, initiation, circulation of material, very boring, very mundane stuff. And we translate that over a series of exercises into something meaningful. On the right hand side we can see a nice visualization that describes what goes on in three sets of processes. So we've got three categories of actions. The first is about tagging music, indexing it, linking it, making sure it's storable and retrievable and that people can find it. That might also mean that you can find it in your own library of sounds. So often this is a self-driven process. There are motivations around putting things in places that you can find them and storing things in ways in which you can access that content. Again going back to that idea of metadata. And then the provenancing stuff, which is all about ownership. There's been a lot in the media recently about the ownership of musical compositions. There's been various legal cases about what it means to create original music. But what we can see from a lot of these life cycles is what that remixing culture looks like, how people adapt and change music, the tools of the trade that they use in engaging in these processes, the signing of the sound, which is a really original process to each composer, to each mixing and mastering engineer. There's a very specific set of activities they engage in, which is personal to them, which is really about branding and marketing. I make music that sounds a particular way. And so I brand myself in that way and there are a series of activities that I engage in in my creative practice that mean that I sign the sound to make it sound like my music, which is a really important process. And then finally the

stuff around legality. So the copywriting of information, intellectual property, which is often defined as IP, and then reusability to make things accessible for other people. So lots of these communities of people we engage with looked at the reusability of their compositions because they want other people to use their component parts in their composition. So through this process, what we find is that we take these really complex, very wordy field reports and translate them into something meaningful. So we produce a meaningful visualization which describes the structure of the component parts of the system, and then that can be translated into something that is for a particular community of people, whether that's a general research community talking about problems in existing systems and ways in which we can improve those systems, or looking for opportunities for design innovation where a new system could come about as a result of these findings. So what we're often doing with trying to visualize information is we're trying to make it more accessible, trying to help everybody to be able to access that information in a meaningful way which is easy to understand and interpret. So when you're producing your research reports, when you're doing research and communicating ideas and concepts in factual ways, in descriptive ways, you want to rely heavily on visualizations. And the two types of visualizations that are really important for you are tabular-based visualizations where you can take a table, an XY representation, mapping one thing against some other things, a really good way to describe complexity in systems, and the other way is through visual structures, what we often refer to as figures. So if you've got a visual way that you can describe a component, an idea, a concept, particularly for interrelated concepts, this can be really useful. Those two types of techniques will really help to bring your research and communication along and help other people to engage with those things in practical ways. So these are the things you want to consider in the work that you're doing when you're trying to produce compelling, rich, interesting visualizations.

Hello and welcome. In this session, we're going to dive into an important concept, design heuristics, and how they can guide the development of better digital audio workstations, or DAWs, to enhance music production workflows. By the end of this video, you'll have a better understanding of how to identify key usability issues in DAWs and how to draft your own heuristics based on research findings. We'll be drawing on a specific example from my own research, which I'll walk you through today. Before we jump into examples, let's take a moment to define what design heuristics are. Design heuristics are broad guidelines that help designers create more usable systems. They aren't rigid rules, but instead flexible principles that can be adapted to different types of systems, including DAWs. They can help you design better systems and help evaluate flaws in existing propositions. Design and evaluation go hand in hand. For example, some classic usability heuristics include consistency and standards, making sure that similar actions are performed in similar ways across the system. Imagine having the maximize window button appear somewhere different every time you open an

application. How annoying would that be? Error prevention, helping users avoid mistakes before they happen. In the context of DAWs, these heuristics can help ensure that the tools musicians and producers use are intuitive, efficient, and supportive of creativity to prevent users getting lost in complexity. You can produce heuristics for pretty much anything. How to drive, how to eat a sandwich, how to make a cup of tea. There's even an ISO standard for that too. It's a set of abstract do's and don'ts for anything you can interact with. DAWs have become essential tools for musicians, producers, and engineers, but as the complexity of these systems increases, usability often suffers. If you want to see some of these tools in action, pop over to Google Images and type in digital audio workstation. You will see some of the most complex and colorful designs in any software system ever produced. It's really difficult to design these things as they have so many features and functions. When we think about music production, especially in a fast-paced or collaborative setting, flow and engagement are key. Usability problems can disrupt creativity, slowing down the process and making it harder to complete projects. That's where heuristics come in. By establishing key design principles, we can help DAW developers create tools that support creativity rather than obstruct it. The challenge, however, is balancing usability with the complexity inherent in music production. As we'll see in the examples from my research, the aim is to avoid overwhelming users with too many options or too much complexity, while still providing the powerful features they need. Whether designing a new DAW or redesigning an old one, there are general lessons to be learned that do not require user input or even user testing. Heuristics are rules-driven approaches, often based on lots of experience or even sometimes from rigorous A-B or multivariate user testing. In my research, I conducted a series of workshops and ran focus groups with professional musicians and producers. The goal was to identify common usability problems they encountered with DAWs and generate ideas for improvements. Notice first how I describe the roles. This is important because we can then start to consider if the system actually works for these types of people. Participants were split into three core roles. Performers – musicians who are primarily focused on live performance or composition. Producers – technologists who work primarily indoors, editing, mixing and arranging music. Agents – stakeholders who support the production process like managers or label representatives. These roles revealed different priorities and usability concerns. For instance, producers wanted efficient workflows, while performers valued flexibility and creative freedom. This makes sense as a live performance is likely to manifest differently to a more deliberate set of tweaks that is required of a mixing or mastering engineer. Let's explore some of the key tables and visuals from the research that show how these findings led to the creation of heuristics for DAWs. One of the first exercises in the research was identifying usability challenges. Too much control. Too many features cluttered the screen and confused the user. Participants wanted a cleaner interface with fewer distractions during creative tasks. Poor workflow. Users struggled to

see the flow of their work, making it difficult to understand where they were in the process of production. Unclear metaphors or symbols. DAWs often used icons and symbols that didn't match real-world instruments, causing confusion. Inconsistent controls. Controls behaved differently across different sections of the software, leading to inefficiencies and frustration. The table is a great example of how we can take real-world usability problems and map them to specific heuristics. For instance, if a system has too many controls cluttering the screen, we can apply the heuristic of efficiency to guide developers towards a more streamlined interface. This actually makes sense when you imagine an analog counterpart. 20 guitar pedals on the floor is easy to visualize and map the flow of. They all light up. They're all interconnected in a linear fashion. Unfortunately, most desktop monitors aren't 15 feet wide, so we can see this problem manifest in real-time. Notice how the table provides a speedy look-up mechanism. After identifying issues, participants were encouraged to suggest solutions based on their workflow challenges. Enabling real-time collaboration and direct commenting within DAWs to streamline communication. Simplify controls with one-click functionality to reduce the user's cognitive load. Improve organization with filtering options to quickly find relevant tracks. These solutions directly inform the design heuristics that were developed. By focusing on problems such as clustered interfaces and inefficient workflows, the participants were able to recommend practical changes that could enhance creativity and usability. A lot of these aren't new features requested as part of the workflow, but some of them are about simplifying repeated processes into something more manageable. Another key finding from the research was the importance of dynamic roles in the music production process. The roles of performer, producer, and agent often shifted throughout a project, and the DAW needed to support these transitions fluidly. One of my future papers actually presents this as a Venn diagram, which is much easier to understand. You could even go further and use something like a concept diagram. You can see examples of these in the paper I wrote with staff at Cambridge, Edinburgh-Napier, and Brighton universities respectively. What this shows is that complex visuals can be presented in accessible ways, a key skill. The heuristic here would be flexibility, ensuring that the DAW can adapt to different roles and workflows seamlessly, without forcing users into a rigid process. Rather than taking a task-based approach, I can flip to a user-centric view that encapsulates any and all requirements of these users. Now it's your turn. Based on what we've explored today, your task is to draft a set of design heuristics that could improve some given system. Try to pick something complex like the DAW example. Maybe an integrated development environment, an IDE with all the bells and whistles, and plugins, would be good. Here are some steps to help you get started. Start with identifying key usability issues. Reflect on the challenges you've encountered in technical systems. What frustrates you? What slows down your workflow? Can you find evidence of these being problematic or design guidelines that suggest best practice? Relate these issues to

heuristics. For each problem, think about which design heuristic might solve it. Is the issue related to efficiency, consistency, or error prevention? Nielsen, Norman, and Schneiderman all produce their own heuristics for general cases, but try to focus on a context-specific domain where the rules might be a bit more nuanced. Try to propose practical solutions. Once you've identified the issues and mapped them to heuristics, suggest improvements. How could the interface or workflow be simplified or enhanced? Sometimes this is as easy as taking a good and bad example and seeing strengths and weaknesses. In conclusion, moving forward with heuristics. Design heuristics provide us with a powerful tool to evaluate and improve complex systems like DAWs. By focusing on core principles like efficiency, learnability, and flexibility, we can create systems that support creativity while maintaining usability. It's essentially a table with possibly some numbers around severity and maybe some colors to determine prevalence. They're incredibly useful though to improve your workflow and make you super efficient. Now it's time for you to draft your own heuristics based on the findings we've discussed and your own experiences. Remember, the goal isn't to oversimplify the system, but to ensure that it's powerful enough to support complex work while being intuitive and efficient. Thank you for joining me today, and I look forward to seeing the heuristics you come up with. Let's make the world a better place, one design guideline at a time. Feel free to dip into other techniques to help visualize this, like the flowcharts we saw, tables of requirements, or something else that helps bring the concept to life.

Welcome back. Now that you've prepared your mobile app pitch presentation, it's time to think about how we'll evaluate both the style and the content of these presentations. Whether you're pitching to investors, a project team, or users, delivering your message clearly and effectively is key. To help guide you, we're going to introduce a set of heuristics to evaluate the quality of your pitch from a communication perspective. Let's start with content. A successful presentation needs to be well-organized and easy to follow. Here are some heuristics to keep in mind. Logical flow. Is your presentation structured in a way that logically takes the audience from the problem to the solution? A good flow might begin with the pain point your app addresses, followed by the key features, and end with the potential impact. Simplicity and focus. Avoid overwhelming your audience with too much detail. Instead, focus on the core features and benefits of your app. Ask yourself, have I clearly articulated the app's main value proposition in a way that's easy to grasp? Audience appropriateness. Tailor your language and depth of detail to your audience. For technical stakeholders, dive into the features. For non-technical audiences, focus on the user experience and business potential. This is so difficult to do for these videos because I don't have an audience. It's much easier to teach in class as I can see faces and look for reactions. You might be falling asleep right now and I wouldn't even know. Tragic. Now let's look at the style of delivery, which is

just as important as the content. Confidence and enthusiasm. Are you speaking with confidence? Enthusiasm is contagious and your audience will be more likely to engage with your idea if you show passion for your app. Remember, you're not just selling the app, you are selling your belief in it. Clear and concise communication. Are you using clear, concise language? Avoid jargon unless it's appropriate for your audience. Be mindful of speaking too quickly or too slowly. Balance is key. I know that I talk too fast and I have to be really conscious of it, especially because I drink so much coffee. Visual aids and support materials. Visual aids should enhance, not distract. Make sure slides or demo materials are clean, focused, and support your key points. Are your visuals helping to clarify your message or are they cluttered and confusing? Have a look at my heuristics or the big poster we presented. That was A1 size but could still be interpreted by different audiences because we had a colourful flow and visuals to help. Audience interaction. Pay attention to how you engage the audience. Are you making eye contact, posing for questions, or checking for understanding? Remember, presenting is a two-way street. You're not just delivering information, you're connecting with your audience. TED Talks are great examples of this. Often abstract concepts are presented in an interesting way to pique curiosity and interest. These lectures aren't really about telling you everything you need to know, more so helping you to navigate complex concepts and access the reading and activities more easily. To summarise, here are the key heuristics for evaluating your presentation. Logical flow and simplicity of content. Confidence, clarity, and visual engagement in delivery. Audience appropriateness and interaction. As you prepare to deliver your pitch, reflect on these areas and consider how you can improve both the content and your delivery style. Good communication is about making your message clear, engaging, and memorable.

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Welcome. In today's session, we'll be exploring nine different communication styles that can help you succeed in various professional scenarios. Whether you're managing a

team, pitching an idea, or navigating a sensitive conversation, the way you communicate can have a significant impact on the outcome. Each style has its strengths, and knowing when and how to use them will elevate your ability to lead, collaborate, and build strong working relationships. Let's dive into these styles, and I'll provide some practical tips on how to apply each one effectively. The first style is firm but fair. This approach is all about being clear, direct, and assertive while maintaining respect and fairness. It's useful when setting expectations or holding others accountable. Be clear and direct. Ensure your instructions are unambiguous. If people don't understand what you want, they won't be able to meet your expectations. So for example, if you need a report by Friday, state exactly that. Maintain respect. Even when being firm, always listen to others' concerns and treat them with empathy. Consistency. Apply your decisions and rules equally across the board. This ensures everyone feels as they are being treated fairly. Constructive feedback. When providing feedback, highlight both areas for improvement and strengths to help the other person grow. Being firm doesn't mean being harsh. It's about clarity and fairness in all interactions. Next, we have the assertive style. Assertiveness is about confidently expressing your needs and opinions while respecting the rights of others. Use I statements. These help you express your viewpoint without placing blame. For example, I need this completed by Monday. Instead of, you always delay the report. Balance. Assertiveness is about finding the middle ground. Be confident without crossing into aggression. Active listening. Acknowledge what others are saying even if you don't agree. It shows you respect their perspective while standing firm on your own. Assertiveness is a key to clear, honest, and respectful communication in any professional environment. In situations which require sensitivity, the empathetic communication style shines. This approach helps you build deeper connections by acknowledging emotions and showing understanding. Active listening. Be fully present in the conversation using verbal and non-verbal cues to show that you are engaged. Acknowledge feelings. Validate the other person's emotions. For example, I understand that this is a challenging situation for you. Provide support. Offer help or solutions that address both the emotional and practical needs of

the person that you're communicating with. Being empathetic helps foster trust and collaboration, especially in challenging or emotional situations. The collaborative style is essential when working in teams or making group decisions. It encourages input from all parties and values collective problem solving. Encourage participation. Invite everyone to share their thoughts and ideas, ensuring that diverse perspectives are considered. Facilitate open discussion. Create an environment where people feel safe to express their views without judgment. Consensus building. Aim for a solution that benefits the majority and is acceptable to the entire group. This style is perfect for situations where buy-in from a team is crucial and you want to ensure everybody feels heard. When time is of the essence or when clear leadership is required, the direct and decisive style is most effective. Be clear. Communicate your decisions without ambiguity or over-explanation. Make it easy for others to understand what needs to be done. Take responsibility. Own your decisions and their outcomes. This builds trust and shows accountability. Be confident. Confidence is key here. Deliver your decisions in a way that reassures others that you're making the right call. This style is ideal for high-pressure environments or situations requiring quick decision-making. The persuasive style is about convincing others to see things from your perspective. It's a blend of logic, emotion, and credibility. Use logic and reasoning. Back up your arguments with facts, data, and a solid line of reasoning. Appeal to emotions. Understand what motivates your audience emotionally and use that to strengthen your message. Build credibility. Establish trust by demonstrating your expertise and reliability so people feel confident in your message. Being persuasive is a powerful tool, especially when pitching ideas or seeking buy-in for a project. The supportive communication style is about motivating and encouraging others, helping them feel capable and valued. Offer encouragement. Recognise achievements and provide positive reinforcement. Be available. Make sure your team knows they can come to you for guidance and support. Provide resources. Offer the tools, advice, or assistance needed to help others succeed. This style is great for fostering a positive and collaborative team environment where people feel motivated to give their best.

Sometimes no single communication style fits all situations. The adaptable style is about flexibility, changing your approach based on the context or the audience. Know your audience. Tailor your communication to the needs and preferences of the people that you're addressing. Be flexible. Shift between being assertive, empathetic, or collaborative as needed. Monitor and adjust. Pay attention to how your message is being received and adjust if necessary. This style is valuable in fast-changing environments or when dealing with a diverse range of people and challenges. Finally, the diplomatic style helps you navigate conflict and sensitive situations with tact and composure. Choose your words carefully. Be mindful of the language you use to avoid escalating conflicts and to maintain positive relationships. Stay neutral. In disagreements, avoid taking sides while working towards a solution that benefits all parties. Build bridges. Look for common ground and work to foster understanding between opposing viewpoints. This style is particularly effective in conflict resolution or when negotiating agreements. In conclusion, there's no one-size-fits-all communication style. The best approach depends on the situation, the audience, and your goals. By developing these nine styles, firm but fair, assertive, empathetic, collaborative, direct and decisive, persuasive, supportive, adaptable, and diplomatic, you'll be better equipped to handle any professional scenario with confidence. Take time to reflect on which styles resonate most with your natural tendencies and practice using different styles in different situations. The more versatile you are in your communication, the more effective you'll be as a leader, as a teammate, and as a professional. Thank you for watching and I look forward to seeing how you apply these styles in your own work.

Hello, I'm Liz Wilkinson, Senior Careers Consultant with the University of London Career Service. Throughout this module, I will be talking to you about how the skills you are learning will help you get hired for the jobs you want and develop your longer term career. I will be interviewing industry professionals, sharing their insights on effective working in technology. And I will also be signposting you to some fantastic resources from the Career Service that can support you on your career's journey. In this lecture, I'm going to outline why assertive communication skills, both written and spoken, matter so much in the development of your career and how you can develop your

communication skills using the free resources of the University of London Career Service. More assertive communication skills will improve your approach at every phase of your job hunting. You will have more informative conversations when networking and information gathering. Your CV and applications are more likely to be shortlisted and your interview performance will be more persuasive. And when you finally get a job offer, you will be well equipped to negotiate a good salary. So for every percent of conscious improvement of your communication skills, you will get a very good return on investment. So why do employers and recruiters care about the skill of communication? They highlight communication skills in job ads because staff with assertive communication skills are essential in delivering good products and results for organisations. Good communicators are more effective in team working, resolving conflict with colleagues, keeping their managers briefed on what they need to know, and dealing effectively with customers, stakeholders, and investors. Assertive communicators improve the financial performance and viability of any organisation. And that is why good communication skills are keenly sought by employers. Let's now unpack the concept of communication skills. I have deliberately chosen the term assertive communication skills because it aligns to some key aspects of effective communication in a career development context. Assertiveness is the middle way between passive communication and aggressive communication. What is considered an assertive tone can vary according to cultural context, but whatever the context, assertive communication has three key elements. One, some understanding of the likely priorities of the reader or audience. What do they care about? What do they worry about? Two, some understanding of the relevant facts and evidence to support what you are arguing. Three, a clear and relevant line of argument to follow and often a clear, positive suggestion of what we should do next to move things forward. Take a moment to think about the people you work and study with. Who are the most assertive communicators? How does their communication style move things forward and how might that make them a more valuable hire? This links into global employability skills as communication is one of the 10 global employability skills which we at the University of London Career Service have identified as being in demand by employers across the globe. Also linked to assertive communication skills is persuasion and negotiation, another of the 10 global employability skills. Both are fundamental to your ability to work well with others and they form a key part of the human advantage in an AI enriched environment. Assertive communication becomes more important in an AI enriched world. Effective use of AI can help you improve the clarity of your arguments as well as spelling and grammar and the University of London Career Service provides you with digital tools which use AI smartly to improve your CVs and applications. However, it still takes a human to think through what would convince another human and in job hunting context there is no one better than you to figure how all your education and professional experience matches the recruiter's requirements. A common recruiter complaint is that

now that AI tools are commonly available they now receive too many bland over-generalised CVs. So even if you use AI to get you started in your draft and improve the targeting of your CV or application statements, you will still need to read it through and add your personalised touch. Let's now look at how assertive communication can improve a CV or application. Think like the recruiter. You start with some thought about what the recruiter is looking for, what they are worrying about and what problem is this job created to solve. What do they need reassurance about? Will the new recruit stay? Can they turn the work around to spec and deadline? Will they fit into the team? Have they got a good work ethic? Backing up your statements with evidence. Think about your evidence of skills, not only your technical skills but also your broader global employability skills that make such a difference to your effectiveness in the workplace. How are you showing the results and the impact of your skills? For application statements the STAR approach can provide a good framework. S stands for a brief description of the situation. T stands for the task that you tackled. A stands for the actions you took. This should be the biggest section with four or five strong verbs describing a series of actions. R stands for results and positive impact on the business. The second R stands for reflection. What did you learn? What would you do differently? Check the clarity and relevance of your targeted argument. When you write a CV or application statement you are making an argument that your experiences and skills, both technical and global employability skills, will make you impactful in this job role in this specific working environment. So, take time to re-read the job description and the organisation's website. Is it obvious from your CV or application that you are applying for this particular job? Have you highlighted your most relevant evidence to make it clear that you are a good match for this job? The University of London Career Service is here to help you. We provide webinars, guidelines and AI enhanced tools to help you hone your CVs and applications. And our global employability skills micro-module has a whole section on technical communication skills to help you develop these skills further. You will find more detail on these resources in the sections that follow.

So my name is Dr Sean McGrath, I'm a lecturer in Computer Science at Goldsmiths University and today I'm going to be talking about my paper, Breaking the Workflow, Design Heuristics to Support the Development of Usable Digital Audio Production Tools. Now the subtitle is Framing Usability Heuristics for Contemporary Purposes, so we're going to go back in time a bit and think about these heuristics in relation to designing modern systems and how they can help and support us in the work that we do in developing things like digital audio workstations, composition tools and tools to support musicians in their general process and their general way of working. So the motivation for this work looks at existing systems and finds a mismatch between them and the expectations of users. So these systems that we use most of the time now in terms of

digital audio workstations, compositional tools, they're very good at functional tasks, they're very good at allowing us to orientate towards a specific set of processes that allow us to produce audio in particular ways. Now the challenges here lie in producing systems that are fit for purpose, i.e. context-driven scenario-oriented activities. So generally speaking, the tools are very good for performing functional tasks, they're not very good at supporting different modalities of working and ensuring that people can work in a collaborative, well-informed, decisive way that allows them to orientate their workflow in an efficient manner. So before we can consider whether these systems are usable, we first have to consider who is a user. So I'll be presenting the results of three workshops today. The first workshop is about understanding the use case, the context of use of these systems, i.e. who's using these systems and how we can define them, and then looking at how the systems break in certain types of contexts. So our first workshop helps us to define users and use cases. We use participatory design to help us to explore context-driven, user-centered design as a way to connect the dots and help us to understand how these systems could fit different use cases. So we asked people working in the music industry, producers, musicians, people working in the periphery of the area to describe their roles and agree on a set of constructs to define these roles. So everybody had to agree on a set of constructs, on a set of use cases, and then we tried to define these in a generalized way. So we have three roles essentially. We have a performer. Now traditionally this might be seen as somebody who plays an instrument or a vocalist. You go and see bands all the time. I do anyway. This is something that most of us are familiar with. Live performance could also be more contemporary in terms of live coding and those sorts of activities. So it's about people making music and the process of making music. We also have producers who, again, this is a fairly familiar role, people with musical knowledge about how to process sound, how to process audio. So these might be people working in mixing, people working in mastering, people who are involved in specific processes in allowing a sound to be digitized and then pushed out to a particular audience of people. So the music is consumed in a particular way. It might be through YouTube videos, it might be pushing things to SoundCloud, or it might be through traditional media, maybe places like Spotify streaming services. There's lots of different places that music can end up. And the producer's job is essentially to take it from this idea and this concept around this performance and then take that and push it down into something which people can actually engage with. And then finally, this role of agency, which is about driving that set of processes. So it's people who sit in the periphery of that area who might have a particular role that's not related to one of these two tasks. So we could think about people at the record label involved in actually allowing this production to happen. We could think about management, external stakeholders, fans could be agents in this role. So people who want to engage with your music as an artist. These people are also important to encapsulate within this set of systems. And of course, the agent relies on

technology for more general tasks. So there's things like communication, time management, planning, scheduling, all these tasks around things like metadata and stuff as well relate to the role of agency. Now what this might look like, looks something like this. So we'll have a technical musician, somebody who makes music in some way. We have a technological person who produces this or allows this to come to fruition through a set of processes. And finally, we have a set of constructs around that. So it might be business focused, monetary focused, people who actually want to drive this and sell the music or allow people to access it through payment systems. And then we can look at these roles in relation to people falling within the remit of one or more of these roles. So you might have session musicians who are focused on making money, of course, but also in terms of actually playing music and being involved in that process. They might not be interested in the digital production side of things. They may be, they may fall into that role too, but generally speaking, you might find a session musician falling somewhere between a technical musical person and somebody focused around the role of agency. And again, we've got different constructs in here. So the independent artists might fall into all three categories. So this is an unusual one where they're interested in making the music, they're the composer. They're also the one digitally producing this through a set of tools and they're driving that set of production. They're engaging with their fans, they're talking to their fans, they're selling music to their fans. So there's a whole set of processes here that one individual might be responsible for. So when we think about these systems in relation to these use cases, we also have to think about these fringe use cases where you might have somebody who's involved in all three sets of processes at different stages in their workflow. So we found here that each role is important. Systems are not particularly well designed for this all-in approach. So where somebody falls within multiple categories, the system doesn't pick apart particularly well and allow them that dynamic workflow. And of course, there's a large body of work around this which doesn't support that. So when we think about encapsulating people in the current system design, systems are at this stage in time designed in a way which allows a set of processes to happen, but it doesn't support dynamic workflows. It doesn't support all the gaps that we have here. So plugins might fill some of the gaps and there are tools like SoundCloud out there to facilitate things like discussion, but again, they don't match this dynamic workflow particularly well. They're not well oriented. They're not designed in a way in which is integrated. And of course, there's lots of fuzzy tasks. So people are doing lots of things that aren't specific to any one of these processes, but are important in bringing about the required sound. So previous work tells us that systems do not work in place. There are challenges with working with systems in terms of things like learnability and adapting to new systems. And again, when we look at adaptive systems, they take into account roles, relationships, workflow, orientation. There's a lot more going on in a music production environment than just the specific set of tasks that allow that to come to fruition from a

digital perspective. There's all these kind of human-centered processes that we don't really encapsulate in systems particularly well. So it might look something like this. You start off with an idea and gradually you move up to a more refined version of something. But what's essentially happening with our systems design is that it supports things like mixing and mastering as we get more refined in terms of what we're trying to produce. We have tools that support those processes, but at the bottom end of this around things like performance, ideation, discovery, creating concepts or themes around sounds, there aren't many tools out there that support those sets of processes, at least not in an integrated way. So when we look at HCI research, we can go back to the 90s and the early 2000s and look at how these sets of tools were useful to help us to design systems that are good for human beings to use. So when we look at the way that digital audio production environments tend to draw from schemas of hardware-based studio environments, that's not really a human-centric way of doing things. We're thinking about, well, these systems work in this physical environment, but there's lots of people nowadays, 17, 18-year-olds coming to audio production environments that have never been in a studio or have never picked up a physical instrument. So that modality of working might not be familiar with the way in which people are used to creating audio from this kind of contemporary perspective. And again, there's lots of different tools available to us now that weren't available 10, 15 years ago. And yet many of the DAW systems have carried on designing systems the way that they were designed back then. So we haven't really got any contemporary tools. So we can look at an example of composition and we can look at two different types of compositions. So we've got some tools here that allow us to do certain things and embed certain working processes so we can change key and things like that. We can define certain musical structures, but even here, there's a difference in how these tools work. So we can see two examples of tools. We have a fairly simplified tool at the bottom and then a more full-featured tool at the top. But even this full-featured tool, those of you who play instruments can probably see examples of where there might be a mismatch. There might be things that aren't embedded within the system and there's no encapsulation of metadata in here. So we can't think about how or why certain processes have come about. It's just a set of technical tools to go from step A to step B. So how do we know what we're building is useful? Well, our aim here is to contextualize things and think about digital technology in relation to how fit for purpose it is. So heuristics provide a nice lightweight way to categorize issues according to a set of constructs. So we might start out with an issue around things like features or flow. There may be a disconnect between the system and what people are seeing. There may be an issue with the control element of things, which is really important for a musician or somebody producing audio. They want to have a certain level of control. Examples might be allowing too much control, for instance, having too many things on a page or things not appearing in the correct order or the order that you'd expect them to appear in, no flow, or again, visibility of system

status, where things are going missing or hidden behind different layers of things. This is something that we can manage in a physical environment. We can lay things out in front of us. But of course we have limited screen real estate with things like DAWs. And so we can't have everything on the screen at once. And we have to think about good ways to manage this. Again, disconnects between metaphors being poorly used, unclear symbolism, or things not matching what you'd expect them to be. And again, accelerators not being uniform, things like shortcuts being different across different systems can create real problems. You're increasing the cognitive load on the user. You're asking them to find things, to discover things, to remember things, to learn a new set of controls, and then remember that. Next time they go into another system, are they going to remember that set of controls or will they be different? So there's all kinds of challenges in modern systems and how they work and these disconnects between what people expect and what the reality is. So you might see something like this where you have, an example might be too much control, too many things on each page, and we can factorize these things according to a set of heuristics. So we can say, well, the first heuristic that we categorize this as is an efficiency problem. That's going to cause people issues, they're going to get lost, confused, it's going to take them time to find things, but there's also sub-factors to that. So it might be about learnability. If this is a system that's new to somebody and there are too many things on the page, they're going to have to find a way to navigate that set of structures. And so that's a problem too. And then again, we can think about this construct of visibility or how many things should we have on a page? How many is enough? How can we diversify this in a way in which people can navigate it? And of course the solution really is simplification, it's providing contextual views. But for each one of these things, there are a number of heuristics that these categories might fall into that we have to think about in terms of coming up with a useful solution and thinking about how we can redesign these systems for purpose. So we might have an issue which is really serious. So it might be something like visibility, which is really serious in the context of efficiency. But there is a real big problem here in that if it breaks your workflow, if it breaks your pattern, as a musician, as somebody who's creative, that's going to create real problems for you. If your cognitive load is focused on how to use the system, you're not going to be able to think about how you're making music and how that production value can come about. So this disconnect is a real issue for some contexts. Again, there are some issues where it's not too much of a problem. So things like the order, you can relearn the order, or you can probably find things relatively quickly, even if they're in an order you don't expect them to be in. But when you think about things like flow and visibility, they can be game breakers. They can be game changers. And that's a problem. So we need to think about how we can design these systems in a way in which we can support these really important modalities of working, but also consider these other constructs. Now what are the general findings of the research? Well, screens do not provide a good basis for

embedded complex input output visualizations. So we can see the example of cables, you know, cables get tangled. You see this all the time in physical environments. It's easy to solve that in a physical environment. In a digital environment, it's very easy to get lost. Simply put, there's often too much going on to make sense of a dynamic set of processes. And so we need a workflow like Photoshop where we've got informed designs based on what you're trying to achieve. So contextually driven designs would really help us in this way. Of course, contemporary systems do not support robust knowledge representation. So things like metadata aren't necessarily embedded in the system. There's not that dynamic discussion, that set of workflows happening. We can't see good evidence of collaboration within these systems. It's all about a set of processes and it's all about coming to a set of sounds or a piece of audio. But it's not about the process and that's a problem. And of course, behavior and controls are inconsistent between systems. So fundamentally, the implications for design are as follows. We need to rethink the way that we design these systems. We must consider task orientation alongside contextually driven perspectives. So we need to think about what people are doing and then designing these systems around a set of processes that allow these things to come to fruition. So thinking about the example of programming, we use version control and content management systems when we're writing code. There's no reason why we couldn't embed these systems in our workflow and think about, well, version control would allow me to impart lots of my knowledge about why I've done things or how I've done things. And then my colleagues could see that if we're working on the same piece of audio, we can actually come to a point where we agree on things through these sets of systems rather than trying to attach other sets of processes to the systems that already exist that allow us to do that. So it's really about thinking about designing systems for human beings and not for these specific sets of processes that we've already really got a good set of designs for.

Welcome. Today we're going to focus on an essential skill for your future in computer science, presenting to technical audiences. Whether you're speaking at a conference, presenting at an industry event, or even delivering a product demonstration, the way you present can make or break how your audience receives your message. Presenting to technical audiences, like your peers, industry professionals, or developers, has its own unique challenges. It's not just about being informative, it's about conveying complex ideas clearly while keeping your audience engaged. In this video, I'm going to share five specific tips for presenting effectively in the field of computer science. Let's get started. Tip number one, know your audience's technical level. One of the most important steps in any presentation is understanding who you are speaking to, but this is especially critical in technical presentations. Your audience tend to vary wildly in terms of their technical expertise. Some may be seasoned developers, while others may only have a basic understanding of the subject area. Adapt your depth of detail. If you're

speaking to a group of highly technical engineers, you can dive deep into specific algorithms, code structures, or software architecture. But if your audience includes managers or non-technical stakeholders, you'll need to zoom out and focus more on higher-level concepts and outcomes. For example, if you're presenting on machine learning algorithms, a highly technical audience might expect you to discuss the inner workings of gradient descent, hyperparameter tuning, or model optimization. However, if your audience includes business professionals or decision makers, they might be more interested in what the algorithm does and how it impacts their business goals. Check assumptions. Never assume that your audience has the same technical background as you. Start by giving a brief context, then build on it with more detail. This way, no one gets lost from the start, and those who are already familiar can quickly re-engage. Tip number two. Use visuals to simplify complexity. In technical presentations, it's common to deal with complex concepts, systems, or data. That's where visuals come in. Well-designed diagrams, charts, and infographics can do wonders in simplifying intricate ideas for your audience. Choose the right visuals. For computer science topics, flowcharts, network diagrams, and code snippets are your best friends. For example, if you're explaining how a distributed system works, a visual showing how data flows between nodes is far more effective than a lengthy verbal explanation. Highlight key points. Don't overload your slides with too much detail. Keep your visuals clean, focused, and designed to emphasize the key points that you're making. For instance, if you're showing the structure of a neural network, you might highlight specific layers like the input or output layers, rather than cluttering the slide with every technical detail. Look at my presentations. You're all very bright, capable university students, but I still keep it simple because you will engage with the concepts more deeply during the activities, the readings, the discussions, and the peer reviews. Live demonstrations. For technical audiences, live coding demos or real-time system walkthroughs can be incredibly engaging, but be cautious. Prepare for potential issues and always have a backup plan in case things go wrong. Remember, the goal is to make the complex understandable. Visuals are not just decoration, they're a powerful tool to aid comprehension. We've already seen some visuals used for both storytelling and to help your understanding of the flow of information. Tip number three. Structure your presentation for clarity and flow. A well-organized presentation is easier to follow, especially for technical topics where it's easy to get lost in the details. Start with the big picture. Begin with a high-level overview of the topic. Explain what problem you're solving or the overarching concept before diving into specifics. For example, if you're talking about a new optimization algorithm, start by explaining why optimization is needed in the first place and what problem your algorithm addresses. Break it down into digestible sections. Use a clear structure like problem, solution, and results to guide your audience through your content. This can be especially helpful when presenting code or technical architectures. For instance, in a presentation on microservices, you

might break it into three parts. An overview of the architecture, the benefits and the challenges, and an in-depth look at the implementation. Transitions are key. Technical presentations often have multiple complex parts. Make sure you clearly transition between sections, giving your audience a moment to refocus and understand what's coming next. Phrases like, now that we've covered the basics, let's look at how this applies to real-world applications. But reorientate the audience. Right now, you know what section I'm on, right? Hint, it's tip number three. A solid structure prevents the audience from feeling overwhelmed and ensures your message comes across clearly. Tip number four. Anticipate and address questions. When presenting to a technical audience, expect detailed questions. In fact, it's a good thing. Questions show that people are engaged and thinking critically about your content. Predict common questions. Before your presentation, think about the kinds of questions you might get. What areas of your presentation might need more clarification? Are there any assumptions you've made that could be challenged? Address potential questions in your content. A great strategy is to weave answers to common questions directly into your presentation. For example, if you're discussing a new software tool, you might preemptively explain why you chose a specific framework or library, rather than waiting for someone to ask the question. Handle questions with confidence. If someone asks something you don't know the answer to, it's okay to say, I'll need to look into that and get back to you. Avoid trying to bluff your way through an answer. It can undermine your credibility. I'll often try things live in a lecture and sometimes get it horribly wrong, but I don't know everything. Understanding the process of discovery is sometimes useful, too. By anticipating questions, you'll feel more confident, and your audience will appreciate that you've thought through the material deeply. Use real-world examples in case studies. Finally, technical audiences want to know how theoretical concepts apply in the real world. Grounding your presentation in real-world examples or case studies is an excellent way to demonstrate the practical impact of your work. Use case studies. For example, if you're presenting on a new data structure, show how it improved performance in a specific application. Explain the real-world problem it solved and the measurable outcomes. Case studies are relatable and help the audience connect dots between theory and practice. People generally like numbers, which leads us to our next point. Demonstrate results. If possible, quantify your success. If your algorithm reduces processing time by 50%, show the before and after comparison. Technical audiences love data, and concrete results give your presentation credibility. Highlight applications in industry. Especially at industry events or conferences, showcasing how your solution fits into broader trends or can be applied across different sectors will captivate your audience. For instance, you could explain how your research on blockchain technology could impact cybersecurity in financial systems. Examples make abstract concepts tangible, helping your audience see the relevance of your work to the real world. Let's recap the five tips for effective technical presentations. Know your audience's technical

level. Adjust your depth of detail based on who you're speaking to. Use visuals to simplify complexity. Leverage visuals, diagrams, and live demonstrations to make complex ideas more accessible. Structure for clarity and flow. Organize your content in a clear, logical way to help your audience follow along. Anticipate and address questions. Prepare for questions by predicting and addressing potential gaps in your presentation. Use real-world examples and case studies. Bring your concepts to life with practical examples and measurable outcomes. Mastering these tips will not only improve the clarity of your message, but also boost your confidence as a presenter. Remember, the key to a great technical presentation is balancing detail with clarity and engagement. Keep practicing and you'll become more comfortable delivering impactful presentations that resonate with your audience.

Hello and welcome. Today we're going to explore how to use visual aids effectively in presentations. We'll take inspiration from Presentation Zen, a widely recognized approach that encourages simple, powerful visuals to enhance your message rather than overshadow it. Whether you're preparing for a conference, pitching an idea, or teaching a concept, visual aids can play a key role in helping your audience understand and engage with your presentation. Over the next few minutes, I'll show you how to use visual aids the right way and I'll point you to some useful resources that can help you master this skill. We've already seen a few concepts before, but we'll dig a bit deeper here. Let's get started. Visual aids are more than just decorations. They're tools for enhancing your message and helping your audience process information. Research has shown that people retain information better when it's delivered both visually and verbally. Let's take a quick look at why visual aids are so important. Clarity. Visuals can simplify complex information. Think about it. Would you rather read a wall of text explaining an algorithm or see a simple flowchart that breaks it down step by step? Engagement. Well-designed visuals can grab and hold your audience's attention. A visually engaging presentation keeps people focused, which helps prevent your message from being lost in a sea of words. Memory retention. People remember 65% of information when it's accompanied by a relevant visual, compared to only 10% when it's not. The right visual can make your presentation stick with your audience long after it's over. Now let's dive into some key principles from Presentation Zen, a philosophy developed by Gar Reynolds. The approach is rooted in simplicity, restraint and naturalness, using visuals to enhance your message, not overwhelm it. Here are the core principles you should keep in mind. Keep it simple. Simplicity is key to effective communication. Too much detail on one slide distracts your audience and dilutes your message. Instead, aim for one main idea per slide. If you need to go into depth, spread the content across multiple slides to avoid clutter. Use high-quality images. Images speak louder than words, but only if they're clear and relevant. Always choose

high-quality visuals that align with your message. Avoid generic or cliché stock photos and opt for visuals that are unique, relatable or that add deeper meaning to your topic. Minimize text. If your slides are crammed with text, your audience will end up reading instead of listening. Use bullet points sparingly and reduce text to key phrases. Better yet, use visual representations like charts, diagrams or icons, instead of words, whenever possible. Six words per line and six lines is a good maximum rule, but less is more. Use contrast to highlight key points. Contrast helps your audience focus on the most important information. For example, if you're presenting a graph, use colour to highlight the trend you want to emphasise, while keeping the rest of the chart more subdued. Storytelling. Engage your audience by using visuals to tell a story. Organise your slides like chapters in a book, guiding your audience through a narrative that connects emotionally and logically. This structure keeps the presentation flowing and ensures your visuals support your overall message. Like here, we have 10 topics that are split around different concepts and ideas. Sometimes we read, other times we listen. Sometimes we dig a little deeper into the concepts. Dipping between levels of abstraction and granularity helps you to see the bigger picture. Now that we've covered the basics, let's look at some common types of visual aids and how to use them effectively in your presentations. Firstly, graphs and charts. What they're good for. They're good at visualising data, at showing trends or comparing information. The best practices for these cases is to keep graphs and charts clean and easy to read. Don't overcrowd them with too much data. If you're showing a line chart, for example, stick to one or two lines to emphasise a specific trend. Always label your axes clearly and highlight key data points with colours or annotations. There are lots of tools online and in books about how to use certain types of graphical representation and which is appropriate for which type of purpose. Diagrams and flowcharts. What they're good for. Diagrams and flowcharts are good at helping you to explain processes, relationships between things or systems as a whole. Best practices. Diagrams and flowcharts help your audience understand complex processes at a glance. Use simple shapes and connectors and limit each diagram to one concept. Avoid overcomplicating the flow with too many branches or options. Focus on clarity. Remember my flowchart visualisation? Images and photos. What they're good for. Adding emotional impact or visual interest to your slides. Best practices. Use images that are directly relevant to your content. For example, if you're discussing user experience design, a visual of an actual user interacting with a product is more powerful than an abstract image. Ensure the image resolution is high enough for large displays. Icons and symbols. What they're good for. Representing concepts quickly and simply. Best practices. Icons are perfect for condensing complex ideas into something visual. Use them to represent key points but don't overuse them. Make sure your audience can easily recognise the icons and avoid using too many different icons on a single slide as this can become confusing. Now that you have a good understanding of the types of visual aids, here are some practical tips

to keep in mind when designing your presentation. Limit animation. While it can be tempting to add flashy transitions and animations, too much can be distracting. Keep animations simple and purposeful. For instance, use animations to introduce bullet points one at a time which helps guide your audience's focus. We'll use a few in this module but not enough for it to become overbearing or even worse, inaccessible. Maintain consistency. Consistency in design creates a professional look. Use the same colour scheme, typeface and layout throughout your presentation. This gives your visuals a cohesive feel and keeps the focus on your message. Leave space for the audience. Don't feel the need to fill every inch of your slide. Leaving white space gives your audience breathing room and keeps the slide from feeling overwhelming. Less is often more when it comes to slide design. There are exceptions to this, for example when working with complex medical scenarios and such, but they are few and far between. To create great visuals you don't need to be a professional designer. There are plenty of tools available that can help you create effective, polished visual aids quickly and easily. Here are a few resources. Canva is a free usable design tool that allows you to create stunning visuals. It offers templates for everything from presentations to infographics. PictoChart is great for creating data-driven visuals like charts and infographics. It's easy to use even if you're not familiar with design. Google Slides for quick and collaborative presentations provides an easy way to integrate visuals, charts and diagrams. This and PowerPoint can be used to create surprisingly rich visualizations. Venngage specializes in infographics and is a great resource for creating visual representations of data, processes or timelines.

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Before we wrap up let's quickly cover some common pitfalls to avoid when using visual aids. Overloading the slide. Too many elements on one slide will overwhelm your audience. Stick to one or two visuals per slide and focus on simplicity. Ignoring accessibility. Ensure that your visuals are accessible to everybody. Use high contrast colors to ensure readability and avoid small text or overly complicated visuals that may be hard to interpret for some audience members. There are automated testing tools and plugins for your browser to help with this if you are unsure. Relying too heavily on visuals. Visual aids should support your narrative not to replace it. Don't read directly from your slides. Use them as a complement to your spoken words. In conclusion, effective use of visual aids can elevate your presentations from good to great. By keeping your visuals simple, clear and relevant, you'll help your audience stay engaged and retain more of the information that you're sharing.

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Remember the key principles from presentations in simplicity, focus and clarity are your guide to creating visuals that truly enhance your message rather than distracting from it.

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9:24
Be sure to explore the tools and resources that we mentioned.

Welcome. In today's session, we're going to cover an important aspect of public speaking and presentations, handling audience questions. Whether you're giving a conference talk, leading a team meeting or delivering a workshop, how you respond to questions can significantly influence how your message is received. Managing audience questions effectively is about more than just knowing the right answer. It's about engaging with your audience, maintaining control of the flow and reinforcing your message. In the next 10 minutes, I'll walk you through techniques and tips to handle audience questions with confidence and professionalism. I'll also point out some references to support these strategies along the way. Let's dive in. Firstly, why are audience questions important? Answering questions gives you the chance to clarify your message. Sometimes, despite your best efforts, parts of your presentation may be unclear. Questions allow you to fill in the gaps and ensure your audience leaves with a solid understanding. That's why we have the tutor forums for clarification of any uncertainties. Engage your audience. Answering questions shows that you're open to interaction, making your presentation feel more dynamic and less like a monologue. I infinitely prefer webinars to live teaching because they tend to be more like a Q&A. These lectures are strange because I'm sat in a room talking to myself. Conversations tend to be far more interesting than monologues, right? Show your expertise. Handling questions effectively is an opportunity to demonstrate your depth of knowledge on the topic. Even if a question challenges you, how you respond can build credibility. As communication expert Carmine Gallo explains in his book, Talk Like Ted, the Q&A section is an opportunity to show that you are knowledgeable and adaptable. Audience interaction increases engagement and connection. A big part of handling questions well is preparing for them ahead of time. Even though you can't predict every question, you can anticipate the most likely ones based upon your content. Review your presentations for gaps or tricky areas. Think about the parts of your presentation that might raise questions. Are there any concepts that could be misunderstood or need more explanation? Prepare clarifications in advance. Even if it's just notes of things you could mention, such as relevant facts, figures or anecdotes that you can talk about. Practice with your peers. One of the best ways to prepare is to rehearse your presentations with colleagues or friends. Encourage them to ask questions. They may come up with queries you hadn't anticipated. We often practice conference presentations in our department before presenting at the conference. Prepare responses for tough questions. Think about potential challenges or criticisms that you might face. Have clear, concise responses ready for any common counter-arguments or difficult topics.

This shows that you've thought deeply about your topic and are ready for scrutiny. Have extra information available. Sometimes it's useful to have additional slides or data ready for in-depth questions. For example, if you're presenting on a technical project, you might prepare back-up slides with technical specifications, code examples or detailed metrics that didn't fit into the main presentation. For my on-campus lectures, I usually have two or three extra resources available if my students are struggling with a concept or an idea. As DeWatt suggests in her book, Illuminate, preparing for questions builds confidence and allows you to respond more effectively under pressure. One of the most important aspects of handling audience questions is listening carefully. Active listening helps you to understand the question fully before you respond. Listen without interrupting. Even if you think you know what the person is going to ask, let them finish. Interrupting can come across as dismissive, and you may miss the nuance of their query. Paraphrase the question back. To ensure you've understood the question correctly, paraphrase it back to the person. For example, if I understand correctly, you're asking how this algorithm handles large datasets in real-time. Is that right? This technique not only clarifies the question, but also shows that you are actively engaged in the conversation. Consider the whole audience. When someone asks a question, repeat it or summarise it for the benefit of the entire audience, especially if the person asking is in the back of the room or if you're speaking to a large crowd. This ensures everyone is on the same page. If you think it's a good question, then say so. That's really interesting because we'll make people feel good and ask more questions. Communication expert Julian Treasure emphasises that active listening is not just about hearing the words, but fully processing the intent and meaning behind the question. Once you've understood the question, it's time to respond. Here are a few key points to keep in mind when answering. Be concise. Avoid long-winded explanations. While it's important to provide enough information, try to keep your response focused. Answer the specific question without drifting off-topic. A focused response shows that you respect your audience's time and are confident in your knowledge. Be honest if you don't know. It's perfectly okay to not have all the answers. If you're unsure of an answer, say so, and offer to follow up later. For example, you can say, that's a great question and I'll need to look into it further. I can follow up with you after the session. This builds trust because it shows you're transparent and open to learning. None of us know everything. Socrates is attributed with the quote, I know that I know nothing. This quote encapsulates the idea that wisdom comes from recognising the limits of one's knowledge. It reflects a philosophical perspective that true understanding begins with humility and the acknowledgement that there is always more to learn. Socrates believed that admitting ignorance is the first step towards gaining wisdom. Link your answer back to your key message. Whenever possible, link your answer to the main points of your presentation. This reinforces your message and keeps the conversation relevant to the overall topic. For example, if the question touches on a peripheral issue, acknowledge the relevance

but steer it back towards the core ideas you want the audience to remember. In his book, Crucial Conversations, Patterson et al. highlight the importance of maintaining clarity and focus during high-stakes questions. Not every question will be easy. Some audience members may ask questions that challenge your assumptions or lead the discussion off-topic. Here's how to handle these situations. Stay calm and composed. When faced with a tough or critical question, remain calm. Don't take it personally and avoid getting defensive. Take a deep breath, pause if needed, and respond professionally. Sometimes a brief pause before answering can help you gather your thoughts. Refocus the conversation if necessary. If a question drifts off-topic or doesn't align with the focus of your presentation, gently steer it back. You might say, that's an interesting point, but for today's discussion we're focusing on X. I'd be happy to discuss your question in more detail after the session. Acknowledge different perspectives. If a question challenges your viewpoint, acknowledge the alternative perspective before offering your response. For example, that's a valid concern and many people share that view. In my experience, however, I've found that. This approach shows that you respect diverse opinions, which can foster a positive exchange. By maintaining control over the direction of the Q&A, you ensure that the session remains productive and aligned with your goals. It is important to remember that during a Q&A session, you're still addressing the entire audience, not just the person asking the question. Make eye contact with everyone. Even though the question may come from one individual, make sure your answer is directed to the whole audience. This keeps everyone engaged and avoids making the Q&A feel like a one-on-one conversation. There's lots of tricks for this, like making the shapes of the alphabet with your eye pattern or looking into all corners of the room and then left to right, right to left. This will depend on the size and shape of the room, but scan slowly. Encourage broader participation. After answering a question, you can invite further input from others. You might ask, does anyone else have a similar question or a different perspective on this? This promotes a more interactive and inclusive discussion. Use questions to highlight key takeaways. If you're asked a question that touches on a critical point from your presentation, use it as an opportunity to reiterate your key message. This reinforces your presentation's main idea and helps drive home the points you want the audience to remember. In the art of public speaking, Stephen Lucas suggests that keeping the broader audience engaged during Q&A sessions is crucial to maintaining the flow and energy of your presentations. As your presentation or Q&A session comes to a close, it's important to Give a clear closing signal. If you're running out of time, give your audience a heads up that your Q&A is about to end. You can say something like, you have time for one last question or let's finish with this final question. One last point I'd like to make. Summarise the key points. After the final question, briefly summarise the key takeaways from your presentation. This helps refocus the audience and leaves them with a strong, cohesive message to remember. Invite further questions. Even though the Q&A session may be

over, let your audience know how they can reach you if they have further questions. This could be through email, social media or after the event. This is a fantastic networking opportunity and can help build broader connection options. By closing the Q&A effectively, you reinforce your authority and leave a positive final impression. So that's seven points. We've covered a lot here and we'll now look at some quiz questions then some practical follow-up activities.

Hello again. In this lecture, we're going to explore how you as computer science students can use your presentation skills to improve your performance in recruitment scenarios, such as interviews or assessment centres. Being able to deliver a persuasive presentation to a mixed technical and non-technical audience is often critical in securing your next role in the technology sector. In this lecture, we will explore strategies to improve your presentation skills, enhancing your performance during these critical job hunting phases. Good presentation skills will improve your performance at various stages of job hunting, from interviews to assessment centres. At the interview stage, these skills will help you structure your answers to interview questions and to appear confident and credible. In many interview scenarios, you will be asked to create and deliver a short presentation on a specific topic. The interviewers may then ask you follow-up questions. In more complex assessment centre scenarios, you may be asked to present solutions to case studies or work on group tasks. In each of these scenarios, your ability to clearly convey ideas, demonstrate problem-solving abilities and communicate well with others is crucial. Strong presentation skills help you showcase your technical expertise and interpersonal abilities. Creating a positive impression on recruiters and giving you an advantage over other candidates. Employers and recruiters value candidates with good presentation skills. They are looking for people who can articulate ideas clearly and work well with others. Presentation skills reflect one's communication abilities, which are vital for teamwork, client and funder interactions and leadership roles. In technology sectors, where complex concepts must often be communicated to non-technical stakeholders, clarity and confidence in presenting are especially valued. Recruiters view strong presentation skills as a sign of professionalism and preparedness. By demonstrating these skills, you show potential employers that you can handle real-world situations and effectively share your expertise, making you a more attractive candidate for their team. The key to success here is mastering both the content and delivery of your presentations. As other content in this topic has demonstrated, it is not enough to have technical knowledge. You must also engage your audience and convey your message effectively. This involves clear structuring, emphasising key points and using visuals to enhance understanding. Practising non-verbal cues like eye contact, posture and gestures can help establish a stronger connection with your listeners. And practice, practice, practice. In particular, practice

with any technology that you are using as part of your presentation. Remember, successful presentations are not just about impressing others, but about ensuring they understand and remember your points. In a recruitment scenario, your presentation skills can be used to demonstrate how you would make a positive impact in the job and as part of the team. Focusing on these aspects helps you communicate better, build confidence and make a memorable impact. In recruitment scenarios for the technology sector, it is crucial to demonstrate your ability to make complex information accessible. Tailor your presentation to your audience, whether they are technical peers or non-technical stakeholders. Sometimes you will be briefed to create your presentation for a particular audience. At other times, you should address your presentation to the mix of roles represented on the recruitment panel. Avoid technical jargon unless you are certain the audience will understand it. Instead, use analogies or visual aids to explain complicated ideas. Spell out acronyms and abbreviations unless they are well known by non-specialists. Keep visuals clear and data focused, emphasising how the information impacts the company or project goals. Practice delivering key points confidently and aim to demonstrate how your solution or insights address real-world problems. This sector values innovative thinking, so it can be impactful to highlight fresh perspectives or novel solutions in your presentations. The University of London global employability skills are fundamental to good presentation skills. Effective presentation demands clear and concise communication, helping you to share complex ideas more easily. Emotional intelligence is important because good presentations require thought about what matters to your audience and how they might react to your points. Additionally, delivering presentation often involves teamwork or peer feedback, enhancing your collaboration skills. In recruitment scenarios, you can harness the power of collaboration to improve your presentations by discussing and rehearsing your approach with friends and mentors. In the technology sector, being able to communicate innovative ideas and problem-solving strategies is essential, as is the ability to connect with a global and diverse audience. Therefore, strong presentation skills play a pivotal role in showcasing these key employability skills. AI can be a powerful tool for refining your presentation skills. You can use AI-based software for grammar and language checks and enhancing your message clarity. AI-driven platforms like ChatGBT can help generate content ideas or can be used for practice question and answer sessions, allowing you to prepare for likely interview questions. AI can also help with presentation timing and provide feedback on tone and pacing. For example, Microsoft PowerPoint has an inbuilt recursal coach. However, using AI critically is essential. Don't rely solely on it. Instead, use AI to assist with your preparation, but use human intelligence to improve your understanding of the audience and ensure the presentation is targeted at the right level for them on this occasion. Focusing on making your delivery as human, engaging and interactive as possible. At the University of London Career Service, we provide both digital tools and human interaction to help you prepare your presentation skills for recruitment scenarios.

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Your approach to presentation skills for interviews will vary depending on whether you are a career starter, a career developer or career changer.

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As a career starter, focus on building confidence and learning the basics of structuring a presentation and delivering within the allocated time and to ensure that you are responding to the task.

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For career developers, work on refining your delivery skills and tailoring your message to align with your growing expertise and shift into a more strategic perspective.

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Career changers should highlight their transferable skills and structure their presentations to highlight links between their previous experience and the new role.

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Regardless of your career stage, careful design, rehearsal and self-reflection are crucial in ensuring that your presentations align with your current career goals and showcase your value effectively. A recruitment short presentation scenario will often be followed by a short question and answer session in which the interviewers ask follow-up questions. Take some time beforehand to think of the questions that might be asked in response to your presentation, perhaps discussing it with a friendly colleague or mentor. In the interview, actively listen to each question and take a moment to formulate your response.

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If necessary, clarify it to ensure understanding. Be concise, focusing on directly answering the query while reinforcing key points from your presentation. If you don't know the answer, be honest and suggest how you would go about finding out.

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Use questions as an opportunity to showcase your global employability skills of complex problem solving and adaptability and resilience.

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Engaging confidently with questions leaves a lasting positive impression and demonstrates your capacity to think quickly under pressure. The University of London Career Service is here to help you. We provide support for both AI and human

optimisation of your interview performance and presentation skills. As a student or recent graduate, you have access to specialist digital tools which provide personalised feedback on your presentation skills displayed in your interview answers. These tools are specially designed to improve your interview performance, so this may be more effective in this task than generally large language models. And we offer webinars on effective interviews and group career coaching on the challenges of interviews and recruitment presentations to help you apply for jobs effectively. You will find more details on these resources in the sections that follow.

Welcome to today's lecture on Computer Ethics and UK Law. Now, before you start yawning, because yes, law and ethics can feel as dry as a desert, let me assure you this is actually one of the most exciting, and dare I say contentious, areas of the computing world. Why? Because ethics and law in computing are constantly evolving, and they're often playing catch-up with technology. Think about it. One minute you're downloading a movie, the next you're unknowingly breaking six different laws and entering a philosophical debate about intellectual property. But before we get ahead of ourselves, let's start by defining the basics. What do we mean by ethics and law? And more importantly, why should you, as a computing professional, care? Ethics is all about the principles that govern what's right and wrong.

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It's subjective, varies by culture, and often has no hard enforcement. In contrast, the law is a set of rules enforced by governments, and if you break them, well, let's just say that you might be swapping your coding skills for a lawyer's business card. Now, here's where it gets messy. Just because something is legal doesn't mean it's ethical. For instance, in the early days of the internet, companies could collect mountains of personal data legally, but was it ethical? That's where things start to get tricky. We've probably gone a bit too far the other way now, because you have to pretty much agree to sign your life away to navigate the web. Still, informed consent is a big part of what we do and important to seek from our users. Let's take a case study you're probably familiar with, the Cambridge Analytica scandal. A company harvested personal data from Facebook users, legally, mind you, under Facebook's terms and conditions, but then used that data for, let's say, questionable purposes, like political influence. The legal landscape was murky, but the ethical backlash was enormous, sparking public outrage and contributing to a much-needed reform of data protection laws, like the UK's Data

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Protection Act 2018 and the infamous GDPR. This brings us to a key question. How do we, as computing professionals, navigate this rapidly changing legal and ethical

landscape? And it is changing. In the UK, the legal framework for digital technology is evolving constantly. We have legislation like the Computer Misuse Act in 1990, which was one of the first attempts to address cybercrime. It criminalises unauthorised access to computer systems, a fancy way of saying, hacking is bad, don't do it. But this law was passed before smartphones, social media and cloud computing even existed. That's right, the law is often playing catch-up with innovation. Back in 1990, the idea of storing data in the cloud sounded like sci-fi. Yet here we are, three decades later, asking whether 30-year-old laws are still relevant in today's tech landscape. Take the Data Protection Act 2018, which incorporates the principles of the GDPR. It governs how personal data is collected, stored and used in the UK. But even this recent law struggles to keep pace with AI systems that can predict our behaviour or algorithms that decide whether you get a loan or even a job interview. You'll cover concepts like this from a technical perspective if you do data science and machine learning at level 6, but here we're interested in the impact it has on human beings. Ethical questions spring up everywhere. Are these decisions fair, transparent and bias-free?

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There are a number of ethical dilemmas that remain in computing. Computing professionals aren't just following laws, they have to think ethically. Let's say you're developing facial recognition software. The law might allow it, but is it ethical if that software has higher error rates for people of colour, as several studies have shown? It's not just enough to say that there are technical limitations of the system when in practice this can unfairly lock people out of systems and cause genuine harm. Access needs to be equitable and these implicit biases can leave people feeling disillusioned and hurt. Computing ethics tackles issues like privacy, security, intellectual property and social responsibility. You'll hear terms like the digital divide, which refers to the gap between those who have access to technology and those who don't. Are we, as developers and engineers, making that gap wider or narrower? Issues such as privacy tend to be culturally influenced. Where some cultures might see age as sensitive information, others are happy to share things like their location and medical history without any concerns.

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Let's consider the impact of intellectual property for a moment. Intellectual property laws protect the rights of creators, whether it's software, music or even that meme you made last night. But how strict should those protections be? For instance, should software patents exist or do they stifle innovation by locking down ideas that others could build on? What if Linus Torvalds decided to make Linux closed source? We all use tools like Tex, the C programming language, Apache, Python, liberal licenses that make them accessible to all. Imagine not being able to read Shakespeare or Mark Twain because

they're prohibitively expensive. Lawrence Lessig, an American legal scholar, once argued that code is law. What he means is that the design of our software, our algorithms, can have as much of an impact on our freedom as the laws passed by Parliament. This is why it's critical for you as computing professionals to understand not just the technical but also the legal and ethical implications of your work. Are you creating tools that empower people or are you building systems that surveil, restrict and control them? Speaking of classic text, I'm often reminded of 1984 when this discussion comes around. Are we really that far away from an Orwellian future? I guess it depends where you live. There are plenty of contentious areas in computing ethics and law, especially in the UK. Think about online piracy. Illegal in the eyes of law, but some argue that copyright laws are outdated and overly restrictive. Or how about data encryption? The UK government has long debated whether law enforcement should have backdoor access to encrypted messages. Ethical? Legal? It depends who you ask. Who should have access? When? For how long? And the debates don't stop there. Look at AI. If an AI makes a mistake, say in diagnosing a patient or deciding an insurance claim, who's legally responsible? The developer? The company? The machine itself? Mester backpropagation algorithm senior? These are questions that UK law is still grappling with and they're likely to be topics of heated debate for years to come. We've seen AIs become racist, sexist and homophobic within a short space of time. What if they became invasive or even violent?

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So, to conclude, when it comes to computer ethics in UK law, the ground is always shifting beneath our feet. But one thing is clear. Understanding the intersection of legal and ethical principles is not just useful, it's essential. Whether you're writing the next great software program or managing a server farm, you'll be navigating this wild frontier of digital law and ethics.

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In fact, we'll talk about server farms later on in the module. And with that, you've taken your first steps into this fascinating, often confusing and sometimes controversial world. In our next session, we'll dive deeper into specific case studies and explore how the UK legal system is responding to these ongoing challenges. Thanks for watching. And remember, just because it's code doesn't mean it's simple.

Welcome to today's lecture, where we'll be diving into one of the most controversial intellectual property disputes in recent tech history, Oracle versus Google. This case is often regarded as a landmark in the discussion around copyright in the digital age, particularly for developers, software engineers, and anyone working in tech. We'll cover the basics of the case, explore the contentious issues, and reflect upon the broader

implications for intellectual property law in the computing industry. Before we jump into the specifics, let's frame the conversation around intellectual property in tech. Intellectual property, or IP, is a set of laws and regulations that grant creators control over their inventions or creations, whether that's a song, a book, or in this case, software code. These laws are intended to protect innovation, but as we'll see, they can sometimes restrict it as well. Intellectual property is crucial to the tech industry, where companies fiercely protect their code, designs, and inventions. But what happens when the boundaries of IP aren't so clear cut? Enter, Oracle versus Google, a case that blurred the lines between functional code and copyright. The story begins in 2010 when Oracle filed a lawsuit against Google, claiming that Google had infringed on Oracle's intellectual property by using parts of Java's Application Programming Interface, or API, in their Android operating system. Java was developed by Sun Microsystems, which Oracle later acquired. Java was designed to be open source, meaning developers could use it freely, but Oracle argued that Google's use of 37 Java API packages to develop Android went beyond the limits of fair use and violated their copyright. Here's the billion-dollar question. Can APIs, which are essentially connectors that allow different pieces of software to interact, be copyrighted? Google said no. Oracle said yes. And that's where the legal fireworks began. This was no small dispute. At the heart of the case was not just a claim of infringement, but a fight over billions of dollars. Android powered by Google was on the rise, and by 2010, it had already become one of the most popular operating systems in the world. Oracle sought $9 billion in damages, arguing that Google had used their APIs to essentially build the backbone of the Android operating system without paying for it. Here's a direct quote from Oracle's legal team to highlight the tension. Oracle took what it wanted, including key Java APIs to accelerate its entrance into the fast-growing mobile phone market. On the flip side, Google argued that APIs were more like keys on a keyboard or rules of grammar, essential building blocks that everyone should be able to use to create new software. If Oracle's claim was upheld, it could have set a precedent that would severely restrict how developers use existing APIs to build new technologies. Let's pause for a second and clarify what an API is, because that's key to understanding this case. You can think of an API as a translator or a bridge that lets two different software programs communicate. It defines methods or procedures that one program can use to ask another to perform specific functions. For example, imagine a website that wants to access Google Maps data. The website uses Google Maps API to request location data, and Google returns the information in a format that the website understands. APIs are everywhere. Without them, most of the apps you use daily wouldn't function. This is why Google argued that APIs should be considered functional tools, not creative works deserving of copyright protection. In the first round of the legal battle in 2012, Google came out on top. The judge ruled that APIs couldn't be copyrighted because they were too functional. The judge famously compared Oracle's arguments to trying to copyright

the arrangement of keys on a typewriter. It's a tool, not a creative expression. But Oracle wasn't about to let the matter rest. They appealed the decision, and the case went to the Federal Circuit's court. Fast forward to 2014, and the Federal Circuit Court of Appeals overturned the initial ruling, siding with Oracle. The court ruled that APIs could be copyrighted, and Google's use of the 37 Java APIs was not automatically protected as fair use. This decision sent shockwaves through the tech community. Many feared it would place a stranglehold on software development. Imagine if every API had to be licensed, making it much harder and more expensive for developers to build upon existing technologies. Here's a quote from a Wired article at the time. If APIs are copyrightable, innovation will slow, and the open-source movement will suffer a major blow. With the Federal Circuit's decision in Oracle's favor, the case was far from over. Google appealed, and the legal battle shifted focus to the question of fair use. The US Supreme Court eventually took the case to decide whether Google's use of the Java APIs fell under the fair use doctrine. Fair use, as you might know, allows for the limited use of copyrighted material without permission from the rights holders, in certain cases like education, criticism, or, as Google argued, transformative uses in a new context. Google claimed that its use of the Java APIs to create Android was transformative because they used it in a new mobile environment. Finally, in 2021, after a decade of litigation, the US Supreme Court ruled in favor of Google, concluding that its use of the Java APIs in Android constituted fair use. The Court recognized that Google had used the Java APIs in a creative and transformative way to build a new platform. In the Court's words, Google's copying of the API was a fair use because it only took what was needed to allow programmers to use their skills in a transformative environment. This decision was hailed as a victory for developers and the broader tech industry, allowing for continued innovation without the fear of every API being locked behind copyright. So, what are the broader implications of this case? First, the ruling reaffirmed the importance of fair use in protecting innovation, especially in the tech space. By siding with Google, the Supreme Court avoided setting a precedent that could have severely restricted developers' ability to build new software using existing tools. However, the case also left some questions unanswered. While APIs in this case were deemed fair use, it's still unclear whether all APIs are immune from copyright, meaning future cases could continue to shape how we understand intellectual property in software. Oracle v. Google was a 10-year legal rollercoaster that not only shaped the future of Android but also set the stage for how intellectual property law intersects with the fast-paced world of software development. The case illustrates the complexities of copyright law in the digital age, how it can both protect innovation and, in some cases, potentially stifle it. As we move forward, it's crucial for developers, legal professionals, and policymakers to stay informed and engaged with these issues. Intellectual property will remain a key battleground as technology continues to evolve. Thanks for tuning in, and in our next session we'll explore other high-stakes IP battles in tech.

Welcome to today's lecture on a topic that has sparked debate for decades in the tech world – software patents. Are they a tool that fosters innovation or do they act as a barrier, slowing down the very progress they're supposed to protect? In this session, we'll unpack what patents in software actually do, examine their contentious role in the industry and dive into real-world examples where patents have both helped and hindered development. Let's start with the basics. A patent grants the holder exclusive rights to an invention, typically for 20 years. This means that no one else can make, use or sell the patented invention without permission. In the context of software, a patent can protect a specific process, functionality or technical method used by a program. Sounds simple enough, right? Well, in reality, software patents are anything but straightforward. One key challenge is that software is, at its core, just code – often a set of instructions or algorithms. Should writing an algorithm really be treated the same as inventing a new type of engine or medicine? And what happens when someone invents something obvious, or that many developers could come up with independently? The history of software patents has not followed a clear process. In the early days of computing, patents weren't commonly applied to software. However, this changed in the 1980s when the US Patent and Trademark Office began issuing patents for certain types of software. One pivotal moment came in 1981, when the Diamond vs. Diacase was decided by the US Supreme Court. The court ruled that a computer-implemented process could be patented, provided it was tied to a specific physical process. This opened the door for more patents in the software space, and by the 1990s, the number of software-related patents had skyrocketed. Now let's consider why patents exist in the first place. The goal of the patent system is to encourage innovation by granting inventors exclusive rights to profit from their work. In return, inventors disclose the details of their invention to the public, contributing to the knowledge pool. In theory, this sounds great. Software companies invest significant time and resources into developing new products, and patents help them recoup those investments. However, the key question is whether this model, designed for physical inventions, really applies to

software development. Many argue that patents are essential for encouraging innovation in software. Large companies spend millions on research and development, and without patents, competitors could simply copy their innovations, without putting in the same level of effort. This would discourage investment in groundbreaking technologies. Take Qualcomm as an example. They hold thousands of patents related to wireless technology, many of which are software-based. Without these patents, they would struggle to justify their massive research and development costs. Proponents of software patents believe that they incentivize businesses to invest in creating cutting-edge software, as demonstrated in the following quote. Patents provide the crucial financial backing for the tech industry's most innovative breakthroughs. Patent Attorney 2015. On the other hand, critics argue that software patents actually stifle innovation more than they promote it. Why? Because, in many cases, patents are issued for ideas that are either obvious or already widely used, leading to patent trolls. Companies that hold patents not to innovate, but to sue other people. In fact, a study by the Electronic Frontier Foundation found that much of the software being patented today is either too broad or too generic, resulting in a legal minefield for small developers. A developer might unwittingly infringe upon a patent for something as simple as a one-click purchase button, and find themselves facing an expensive lawsuit. These criticisms are supported by the following quote. Software patents are a mess. They're designed to protect ideas, but they're often just weapons in a corporate arms race. Wired. 2017.

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One of the most famous examples of how software patents can turn contentious is the long-running legal battle between Apple and Samsung. At the heart of the case was Apple's claim that Samsung had infringed on several of its software patents related to the iPhone's user interface, including the infamous bounce-back effect when scrolling. Apple won a $1 billion judgement against Samsung in 2012. A clear victory for the pro-patent camp, but others saw it differently. Critics argued that these software

features, such as tap-to-zoom and rubber-banding, were simple, intuitive ideas that shouldn't have been patentable in the first place. This case highlights how patenting basic software functionalities can hurt competition and make it harder for smaller companies to innovate without fear of litigation. Another huge issue with software patents is the rise of patent trolls – companies or individuals who hold patents not to create products, but to sue others who inadvertently infringe upon their claims. These trolls buy up vague or overly broad patents and wait for someone to use a similar idea in their software before filing a lawsuit. One famous example is Unilock, a company that has sued Microsoft, Google, Apple and many others for infringing upon software patents it holds. Critics argue that this type of behaviour discourages smaller developers and start-ups from innovating because they fear being targeted by patent trolls. Given these issues, some have proposed alternatives to software patents. One approach is to use open-source licences, where software code is made freely available to the public. This model encourages collaboration and shared innovation without the legal barriers that come with patents. For example, Linux, one of the world's most widely used operating systems, operates under an open-source licence. Thousands of developers contribute to Linux without the need for patents, demonstrating that innovation can thrive without restrictive IP rights. It's important to note that different countries approach software patents differently. In the United States, the Supreme Court's decision in AliceCorp v. CLS Bank in 2014 made it harder to patent abstract software ideas. The court ruled that simply implementing an abstract idea on a computer does not qualify for a patent, a decision that has led to the invalidation of thousands of existing software patents. However, other countries like Japan and Germany take a more lenient approach, allowing broader patenting of software. This creates an interesting international dilemma. Should there be a unified global standard for software patents? And if so, what should it look like? So where do we land on the question of software patents? The truth is, they can be both a force for innovation and a barrier to progress. It all depends on how they're applied. When used correctly, patents can protect developers and encourage companies to invest in groundbreaking technologies. But when patents are

too broad, or when they're used by trolls to stifle competition, they become a hindrance. The real challenge for the future of software patents is striking the right balance, ensuring that we protect genuine innovation while fostering a landscape where creativity and collaboration can flourish.

Welcome to today's lecture, where we're focusing on one of the most fun yet contentious areas of intellectual property – copyright laws in digital media. Whether you're streaming the latest album, binge-watching a series, or downloading digital art, copyright has its hands all over it. But copyright isn't just about protecting artists. It's also generated battles over piracy, streaming rights, and who gets to control creativity in the digital world. So grab your virtual popcorn, and let's unpack how copyright laws impact everything from YouTube videos to your favourite means.

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Let's start with the basics. Copyright gives creators exclusive rights over their original work, whether that's a song, a book, a film, or even software code. These rights can include controlling who can copy, distribute, and adapt the work. Copyright protection typically lasts for the lifetime of the creator, plus 70 years in most countries. But here's where things get more interesting. In the digital age, anyone with an internet connection can copy and share works in seconds. This raises questions about how copyright laws, written in the pre-digital era, apply to today's streaming services, social media, and peer-to-peer file sharing. Now, who hasn't heard of digital piracy? It's become a defining copyright issue of the 21st century. From Napster in the late 90s to torrent sites today, digital piracy has shaken up the entertainment industry. In the early 2000s, services like Napster made it possible for users to share music files without paying artists or record labels. Napster's downfall came swiftly when Metallica sued, marking one of the most famous copyright cases of the internet age. Lars Ulrich, Metallica's drummer, famously said, Every time a Napster enthusiast downloads a song, it takes bread out of the mouths of a musician and their family. Of course, fans didn't see it that way, and Napster was just the tip of the iceberg. Other file sharing platforms like Kazaa and BitTorrent continued to thrive, leaving artists and the legal system scrambling to catch up. Let's talk about how copyright has impacted the music industry, which arguably took the hardest hit from digital piracy. A 2007 study by the Institute for Policy Innovation estimated that global music piracy cost the US about $12.5 billion per year. But was it all doom and gloom? In response, the industry had to evolve. Enter streaming services like Spotify and Apple Music. These platforms have been praised for legalising access to music while compensating artists, though not without criticism. Taylor Swift, for instance, pulled her catalogue from Spotify in 2014, arguing that artists were being underpaid for

streams. I'm not willing to contribute my life's work to an experiment that I don't feel fairly compensates the creators, were her thoughts. While streaming has been seen as a solution to piracy, it's sparked a new debate about whether artists are paid fairly by these platforms. The rise of streaming platforms hasn't just changed how we listen to music. It's affected film, TV and even gaming. Platforms like Netflix, YouTube and Twitch are governed by copyright laws, but not without controversy. Netflix and other subscription services pay licensing fees to stream content legally, but YouTube in particular has faced copyright battles because of its user-generated content. Creators are often caught in the middle. YouTube uses an automated system called Content ID, which allows right-holders to claim and monetise videos using their work. However, this system has been criticised for unfairly penalising creators for brief or incidental uses of copyrighted content. A funny meme or a fan edit can quickly become a legal problem. Here's a quote from YouTuber Lindsay Ellis, who's experienced copyright strikes first-hand. You don't even know what will trigger a copyright claim anymore. It's like playing Russian roulette with your content. This highlights how copyright law in its current form is struggling to balance protection with creativity in the digital world. One of the most contentious aspects of copyright is how it affects creative control. Copyright law was designed to protect creators, but it can sometimes be a double-edged sword. For example, fan fiction and remixes are technically unauthorised uses of copyrighted works, but they've become a significant part of online culture. Yet, under strict copyright laws, these derivative works could be taken down. This has led to a tension between copyright holders who want to control how their works are used, and fans who see these as creative expressions, as a form of tribute or free advertising. This tension has been dubbed the Remix Culture Battle. Lawrence Lessig, a leading voice in copyright form, said, We live in a world where the law demands the permission of the past to build the future. What Lessig means is that creators should be able to build on existing works to create something new without facing legal obstacles. Platforms like SoundCloud and TikTok have embraced this remix culture by negotiating licensing deals that allow users to legally create derivative works. Of course, no discussion of copyright in digital media is complete without touching on Fair Use, a legal doctrine that allows limited use of copyrighted material without permission in specific cases like commentary, criticism, or parody. Fair Use has become a vital tool for YouTubers, Twitch streamers, and critics. For instance, film critics often rely on Fair Use to include movie clips in their reviews. But the boundaries of Fair Use aren't always clear, and many creators find themselves on the wrong side of copyright strikes. As one YouTuber, Hbomberguy, humorously said, I live in a constant fear that I'll wake up one morning to find out I've been sued for using a three-second clip of Spongebob. The vagueness of Fair Use makes it tricky for creators to know when they're crossing the line. The laws around Fair Use are still evolving, especially as platforms like YouTube introduce stricter copyright filters. Copyright laws vary greatly across the globe, making things even more complicated for

digital media. For instance, the European Union's Article 17, previously known as Article 13, was designed to increase copyright enforcement online. It requires platforms like YouTube to filter out copyrighted content unless permission from the rights holder is obtained. Critics argue this law is too strict and could stifle free speech and creativity. The Electronic Frontier Foundation described it as a massive overhaul of internet freedoms that prioritizes big copyright holders over individual creators. This is a significant shift, and it's a stark contrast to US Fair Use, which tends to be more lenient. So as the internet continues to transcend national borders, the clash of different copyright regulations is becoming more pronounced. So what does the future hold for copyright in the age of digital media? The industry is clearly in a state of flux. While new laws like Article 17 are being introduced, platforms are also adapting, trying to balance the rights of creators and copyright holders with the evolving landscape of digital culture. We're seeing moves towards copyright reform that could reshape how we approach intellectual property in the digital world. Concepts like Creative Commons licensing have emerged, allowing creators to share their work more freely, while retaining some control over how it's used. As technology continues to advance, copyright laws will need to evolve to keep up with the changing nature of media consumption. In conclusion, copyright law is a necessary but contentious part of the digital media landscape. It exists to protect creators, but in an era of streaming, remix culture, and global connectivity, these laws can sometimes feel like they're struggling to catch up. Whether you're a content creator or a consumer, understanding how copyright works and how it impacts digital media is crucial in today's world. As we've seen, the stakes are high, and the debates are far from over. Thanks for tuning in, and in our next lecture we'll take a deeper look into how these laws are evolving in real time.

Welcome to today's lecture, where we'll be exploring some fascinating and challenging aspects of using adaptive music in everyday journeys. Specifically, we're going to dig into a real study that used GeoTracks, a system that adapts music to the listener's environment based on their location. This study has some serious ethical considerations, especially regarding safety, awareness of surroundings and how to keep music enjoyable without compromising safety. So let's get into it. Let's start with what the GeoTrack system actually does. The idea is to adapt music dynamically as you walk through different environments. For example, if you're walking along a busy street, the music might become quieter and less complex, allowing you to stay aware of traffic and other potential hazards. The study compared two experiences, listening to a regular, fixed playlist versus listening to music that changes based on where you are and what's around you using GPS data. Sounds cool, right? Well it is, but it also raises some ethical concerns. One of the biggest ethical questions in this study was, how do we keep people safe while they're listening to their favourite tracks? We know music can be immersive, sometimes too immersive. Blocking out the sounds of the real world like

approaching cars, cyclists or other pedestrians can be dangerous. To address this, the study experimented with jawbone headphones. Unlike regular earbuds, these headphones transmit sound through the bones near your ears, keeping your ear canals clear so you can still hear environmental sounds. This setup meant participants could listen to music without completely tuning out the world around them, helping them stay aware of potential hazards. As one participant commented, it was nice to still hear what was going on around me without the music cutting me off from the environment. A major feature of the GeoTrack system was its ability to dynamically adapt the music based on where the user was walking. For example, when a participant approached a busy road or a pedestrian crossing, the music would automatically become quieter and simpler. Imagine you're vibing to an energetic song with lots of layers, drums, vocals, guitar and suddenly, as you near a crossing, the system fades most of the instruments, leaving just a soft piano melody. This gives you mental space to focus on the potential danger, like cars or bikes whizzing by, while still enjoying the music. This approach reflects an important balance between entertainment and safety. The idea was to prevent the music from becoming a distraction when users needed to be most alert. Adjusting the music when near hazards may sound like a simple concept, but there are deeper ethical concerns here. How much can you simplify the music before it stops being enjoyable? How do you ensure people still feel the emotional connection to their music without risking their safety? In this study, we tried different approaches. For example, near a busy road, the mix might lower the volume of vocals and drums, but leave instrumental layers intact to keep the mood. However, as noted in the study, some participants were frustrated when their favourite tracks were cut off or dramatically altered. One participant said, I felt like the music didn't flow anymore when it suddenly changed. I didn't like when my favourite part of the song got skipped. This points to a key ethical dilemma. How do we keep music adaptive without ruining the experience? Here's another interesting ethical question. How much control should users have over this adaptive system? Should they be able to override the safety features, or does that put them at greater risk? In this study, the music was adapted automatically and users didn't have control over when or how the tracks changed. While this ensured that participants stayed aware of their surroundings, some participants mentioned that they would have preferred more control over the music, particularly when it came to their favourite songs being altered. This leads to the broader ethical question in tech design. Do we prioritise user autonomy, or do we prioritise their safety, even if it means taking away some control? Do we always know best? Participants gave varied feedback about the experience. Most people agreed that the adaptive music enhanced their walk, making it feel like a soundtrack to their journey. However, some raised concerns when the music didn't perfectly sync with their environment. Here's a direct quote from the study. When the music fit the environment, it really complemented the experience, but when it didn't, it had the opposite effect, breaking the immersion. This suggests a key challenge for

developers of adaptive systems. How do you ensure the music always fits while maintaining users' safety? The system needs to balance technical precision with the unpredictability of real-world environments. A classic case was one participant crossing the road at a time an emergency vehicle approached quickly. Although the music had become softer, the cognitive load of the participant at this point was high, as they were trying to navigate unfamiliar territory. The emergency vehicle created a risk that we could not accommodate for, and might have been present in any scenario where a person is struggling to find their way – tired, distracted, etc. Even though we gave participants a map and talked them through the walk, they did not always necessarily understand the route. Fortunately, ambulance drivers are well-trained in driving at speed, and they know to exercise caution in these scenarios. As a lesson, we might ask participants to do the walk without music first, so that they are familiar with the route. Let's not forget the privacy concerns in this type of system. Since GeoTracks relies on GPS tracking, it collects real-time location data about its users. While the study didn't focus heavily on privacy, any system that tracks location raises questions. Who has access to this data? How long is it stored? What are the potential risks of data breaches? Ethically, developers must ensure that all data is anonymised, stored securely, and used responsibly. Users should also be fully informed about what data is being collected and why. We also took interview data, which I then transcribed, coded to remove people's names, and stored on a USB drive that was locked away in one of the professor's offices. So to recap, the main ethical challenges in this study involved balancing safety, user enjoyment, and control. The use of jawbone headphones ensured participants could still hear their surroundings, promoting safety while listening to music. The system adapted the music dynamically based on the environment, such as quieting it near busy roads. There was a trade-off between giving users a smooth, enjoyable experience and ensuring their safety, which sometimes led to frustration. And finally, we must always consider privacy issues in systems that rely on GPS data. This study is a great example of how technology can enrich our everyday experiences, but also how it introduces new ethical considerations that developers must carefully navigate. Adaptive systems like GeoTracks offer a promising future for enhancing everyday experiences, but they come with a unique set of ethical challenges. Whether it's balancing safety with enjoyment, managing data privacy, or determining how much control to give to users, these are critical questions for anyone designing tech that interacts with real-world environments. Thanks for joining me today to unpack these ethical issues. You can go ahead and read the paper if you wish. It was fun to write, and even more fun to run the study, though full of technical challenges.

Welcome to today's talk on sustainable computing practices, a topic that's becoming increasingly urgent as our digital world grows at lightning speed. We're going to look at how we balance the ever-growing need for computational power against the very real

environmental costs that come with it. From AI systems crunching massive data sets to streaming services delivering billions of hours of content daily, computing is thirsty for energy. But as we race to build faster, smarter machines, we face a trade-off. How do we power the future without scorching the planet? Today we'll explore this trade-off, examine some sustainable practices in computing and dive into a case study that illustrates the delicate balance between tech innovation and sustainability. Let's start by highlighting the core issue. Computational power is expensive, and I don't just mean financially. Every calculation, every query, every video streamed has an environmental cost. Data centers, the backbone of our cloud computing and digital world, use around 200 terawatt hours of energy annually. To put that into perspective, that's almost the same amount of electricity used by Argentina. And let's not forget the cooling systems needed to keep these centers from overheating, which adds to the power consumption. According to a Gartner report, by 2040, the ICT sector, information and communication technologies, could use 14% of the world's energy. And that includes everything from data centers to your smartphone. The environmental toll doesn't stop at electricity use. Data centers generate around 2% of global CO2 emissions. That's on par with the aviation industry. Let's think about that for a second. Every time you ask Siri a question, backend servers fire up to retrieve data requiring energy, which in turn produces greenhouse gases. Now, as the demand for AI systems grows, and as cloud services scale, this carbon footprint could skyrocket. Consider this, training just one large AI model like GPT-3 can emit as much CO2 as five cars over their entire lifetimes. That's not a small impact. So here's the dilemma. On one hand, we need computing power for everything, scientific discoveries, medical research, entertainment, social media. But on the other hand, that power comes with a heavy environmental cost. This raises some difficult questions. Should we limit computational growth? Can we innovate in a way that reduces our carbon footprint while still meeting the demand for more power? And most importantly, how do we balance these competing needs? Fortunately, the tech industry isn't ignoring these concerns. Many companies and researchers are working on sustainable computing practices to balance power demand with environmental concerns. Here are a few approaches making headway. Green data centers. By using renewable energy sources like solar and wind, data centers can reduce their reliance on fossil fuels. For example, Google and Microsoft have committed to running their data centers on 100% renewable energy. Energy efficient hardware. Companies are developing more energy efficient chips and processors like ARM architecture chips which use far less power than traditional processors. Dynamic resource allocation. This involves optimizing the use of computing resources so power is only consumed when absolutely necessary. For example, serverless computing platforms scale automatically and use power only when a service is running. Carbon offsetting. Some companies are purchasing carbon credits to offset the emissions they can't avoid. While not a perfect solution, it's a step in the right direction. Let's now get into real world examples of this

trade-off. Cryptocurrency mining. Cryptocurrencies like Bitcoin require vast amounts of computational power. Every Bitcoin transaction is verified by a process called mining which involves solving complex mathematical puzzles. As the puzzles become more difficult, miners need more powerful hardware which consumes huge amounts of energy. To give you an idea, Bitcoin mining alone consumes more electricity annually than the entire country of Sweden. This has raised massive ethical and environmental concerns. But here's the interesting part. Some crypto companies are trying to go green. Projects like Chia and Cardano are developing blockchain technologies that require less energy, while others are moving towards using renewable energy for mining operations. There's a clear trade-off here between the innovation cryptocurrency offers and the environmental cost it incurs. Let's take a closer look at Chia, a cryptocurrency designed to be more environmentally friendly. Instead of using the energy-intensive process of Bitcoin mining, Chia uses a system called Proof of Space and Time. In simple terms, this method uses unused hard drive space to validate transactions, significantly reducing energy consumption. Sounds good, right? Well, here's where the trade-off comes in. While Chia consumes less electricity, it places wear and tear on hard drives, causing them to wear out faster. This creates a different environmental issue – e-waste – so even when we try to make greener systems, we still face new challenges. AI can be both part of the problem and part of the solution in sustainable computing. In large AI models, as I mentioned earlier, consumes enormous amounts of energy. But AI also offers opportunities for optimisation. For example, AI-driven algorithms can help improve energy efficiency in data centres. Google, for instance, uses AI to optimise cooling systems in their data centres, reducing energy use by up to 40%. That's a significant drop in power consumption, showing that AI can be leveraged to solve the very problems it creates. Let's now take a look at another example – machine learning in climate science. While machine learning models require significant computational power, they also help us understand and fight climate change. For instance, researchers are using machine learning to predict deforestation patterns, improve weather forecasting, and even optimise renewable energy grids. This is a great example of how computational power can be directed toward environmental benefits, highlighting the trade-off in a positive light. So where do we go from here? Sustainable computing is still a developing field, but there are several steps we can take to reduce the environmental impact of our digital lives. Adopt renewable energy. The most straightforward way to reduce the carbon footprint of computing is to power data centres and devices using renewable energy sources. Improve hardware efficiency. We need to continue developing energy-efficient hardware, which will reduce the energy consumption of every calculation, whether in AI, cloud computing, or cryptocurrency. Educate developers and users. Tech companies and users alike must be aware of the environmental impact of their choices. From designing more efficient algorithms to turning off idle servers, every little bit helps. We also can't overlook the ethical and

social implications. As we push forward with advanced computing technologies, we must ask ourselves. Is it ethical to prioritise computational power over environmental preservation? How can we ensure that advancements in computing benefit everybody, not just those who can afford more efficient, cleaner technologies? In the end, the goal is to find a balance between progress and sustainability, a balance that requires thoughtful consideration of not just what we build, but how we build it. In conclusion, sustainable computing is a tricky balance between power and planet. While we need ever greater computational power to fuel innovation, we also have to take responsibility for the environmental costs. As we've seen through case studies like cryptocurrency and AI, there are no perfect solutions, only trade-offs. But with continued efforts in green data centres, energy efficient hardware and AI optimisation, we can work toward a future where technology and sustainability go hand in hand. Thank you for joining me today. In our next session, we'll explore how edge computing could help reduce our reliance on energy-hungry data centres.

Welcome back to another deep dive into sustainable computing. Today we're going to explore some new approaches that balance the increasing need for computational power with the urgency of addressing our environmental responsibilities. As we all know, the demand for digital services is skyrocketing, but so is the environmental impact. Today's session is packed with data and fresh examples of how the tech industry is working to build a greener digital world. So let's jump in. Let's start by looking at the growing need for computing power. Global data consumption is expected to double every two years, with over 175 zettabytes of data predicted by 2025. Streaming services, cloud computing and AI are all major contributors to this surge. With this explosion in data comes an increase in energy demand. Currently the information and communication technology sector accounts for around 4% of global electricity usage, and that number is set to rise. As tech giants and industries push the boundaries of what's possible, it's become increasingly clear that sustainable solutions are not just an option, they're essential. Here's a staggering statistic. Data centres alone produce roughly 3% of global CO2 emissions. That's nearly the same as the airline industry. It's not just about energy usage, but also water consumption. Cooling systems in data centres guzzle millions of litres of water to keep servers from overheating. In 2020, Microsoft's data centres in Quincy, Washington consumed over 250 million litres of water in a single year. With more data centres popping up worldwide to handle increasing demands, the environmental footprint continues to grow. This brings us to the heart of today's discussion, what can be done? To tackle these challenges, several sustainable computing strategies are emerging across industries. Today we'll be focusing on three big themes. Green AI, making artificial intelligence models more efficient. Cloud optimisation, reducing the carbon footprint of cloud computing. Circular computing, extending the lifecycle of hardware through recycling and refurbishing. Each

of these practices offers unique solutions to the growing conflict between computational power and environmental sustainability. Let's start with AI. As we train more complex AI models, the energy demands increase exponentially. A 2019 study showed that training a large language model like GPT-3 can emit 284 tonnes of CO2, which is equivalent to the carbon footprint of five cars over their entire lifetimes. That's just one model. Researchers are starting to develop green AI approaches, which aim to reduce the energy needed for training. One approach is to focus on efficiency, using smaller models that achieve similar accuracy but require significantly less computational power. Another is to adopt transfer learning, which involves training AI models on smaller datasets first and gradually refining them, thus cutting down the energy and time required. One notable example of green AI in action is NVIDIA's energy-efficient AI solutions. NVIDIA has introduced hardware optimised for AI model training with reduced energy consumption. For example, their DGX-A100 systems can deliver significant performance improvements while reducing overall power consumption by up to 30%. By pairing hardware improvements with software optimisations, NVIDIA is showing how cutting-edge AI technology can become more sustainable. It's a big step in making AI innovation cleaner without sacrificing performance. Next, let's talk about cloud optimisation. The rise of cloud computing has created vast networks of data centres around the world, but they come with high environmental costs. In response, companies are looking for ways to optimise cloud services to reduce energy consumption. One approach is dynamic resource scaling, which automatically adjusts server capacity based on real-time demand. This prevents data centres from running at full capacity when they aren't needed. Another innovation is moving to edge computing, processing data closer to the source rather than in centralised data centres. This reduces both latency and energy use, especially for IoT devices. And let's not forget renewable energy. Companies like Amazon Web Services and Google Cloud have committed to powering their data centres entirely with renewable energy. Google Cloud, for example, has already achieved carbon neutrality and aims to operate completely on carbon-free energy by 2030. Let's delve deeper into Google Cloud's commitment. Google has been at the forefront of sustainability and in 2020 they became the first major company to operate their data centres at 100% carbon neutrality. But they're not stopping there. They've committed to using carbon-free energy, wind, solar and geothermal for all their operations by 2030. Their machine learning models now predict demand spikes, enabling data centres to adjust their power usage in real-time. This has resulted in a 40% reduction in the energy needed to cool their data centres. It's a shining example of how tech innovation can work hand-in-hand with sustainability. Now let's talk about hardware. As computing needs grow, so does the demand for more powerful processors, servers and devices. But constantly manufacturing new hardware creates a huge amount of e-waste. In 2021 alone, the world generated 57 million metric tonnes of e-waste, the equivalent of about 4,500 Eiffel Towers. Circular computing aims to solve

this by extending the lifecycle of hardware through recycling, refurbishing and repurposing. Instead of throwing away old servers and devices, companies are refurbishing them, ensuring they have a second or even a third life before ending up in landfills. This approach not only reduces waste but also cuts down on the carbon footprint involved in manufacturing new hardware. One excellent example of circular computing in action is Dell's commitment to sustainable hardware. Dell has introduced programs that reclaim and refurbish old hardware, turning them into new products that can be reused by businesses. Their Optiplex 7070 Ultra series is a fully modular desktop system made from recycled plastics. The company aims to ensure that 100% of their packaging will be made from sustainable materials by 2030. Dell's efforts in circular computing demonstrate how tech companies can reduce both waste and the energy needed for new product manufacturing. But here's the challenge. While circular computing is a great step forward, the global demand for new devices continues to rise. This creates a dilemma. How do we curb e-waste when consumers demand the latest tech? Tech companies must not only ramp up their recycling efforts but also consider how they design hardware to be more repairable and reusable. Take Apple for example. Their newer devices, while sleek and high performing, have been criticized for being difficult to repair. Moving toward modular designs that can be upgraded rather than replaced entirely could be key in reducing e-waste. Now that we've covered a range of sustainable computing practices, let's talk about what we can do as consumers, developers and businesses. Use energy efficient solutions. Whether it's using energy efficient cloud platforms or opting for devices that consume less power, every choice adds up. Opt for refurbished tech, when possible, buy refurbished or recycled tech products rather than new ones. This helps reduce e-waste and promotes circular computing. Optimize your code. Developers can write more efficient code that reduces the computational power needed to run applications. Think of it as going green at the coding level. Support sustainable companies. Lastly, we can support companies that are committed to reducing their environmental impact, whether it's through sustainable cloud services or eco-friendly hardware. In conclusion, the path to greener computing isn't easy, but it's essential. Whether it's reducing the energy demands of AI models, improving the efficiency of cloud services or extending the life of our hardware through circular computing, there are solutions out there. As demand for computational power continues to rise, we'll need to innovate and adopt sustainable practices that can keep up with technological advancements while minimizing their environmental cost. Thanks for joining me today and remember, every small step towards sustainability counts in our digital world.

Welcome to today's lecture on handling legal risks in IT projects. As we know, technology is an essential part of every business today and IT projects are more ambitious than ever. But with the ambition comes risk, legal risk to be precise. Whether

it's handling contracts, data protection, intellectual property or compliance, legal risks are always lurking in the background. Today we're going to explore these risks, look at a few real-world case studies and discuss how to avoid the pitfalls that can derail even the best laid IT plans. Let's begin by defining what we mean by legal risks in IT projects. Legal risks refer to any potential issues that could lead to financial loss, legal disputes, regulatory fines or damage to reputation due to failure to comply with laws and regulations. These risks come in various forms, such as breach of contract, intellectual property violations, data protection breaches, non-compliance with regulations. As IT projects often involve multiple stakeholders, complex technologies and sensitive data, legal risks can appear at any stage from project initiation to deployment and beyond. Here are some of the most common legal risks encountered in IT projects. Contract disputes. Unclear or poorly drafted contracts can lead to disagreements over project scope, deliverables or timelines. These disputes often end up in court, causing costly delays. Data privacy and security. With data breaches becoming more common, ensuring compliance with regulations like the GDPR – General Data Protection Regulation – is critical. Failure to protect personal data can result in hefty fines and a loss of trust. Intellectual property or IP issues. IT projects often involve the creation of software, algorithms or other intellectual property. Without proper agreements in place, disputes can arise over who owns the IP. Compliance with industry standards. Many sectors such as healthcare and finance are subject to strict IT regulations. Non-compliance can result in severe penalties. We'll look at a few case studies where these risks have caused major headaches. Our first case study is one of the most infamous IT project failures in the UK – the NHS National Programme for IT. This ambitious project was launched in 2002 with the goal of creating a centralised digital healthcare system for the NHS. The project quickly spiralled out of control, facing technical difficulties, delays and legal disputes. One of the key legal issues in this case involved contract disputes. Several contractors, including BT and CSC, were brought on to deliver various aspects of the system. However, unclear contracts led to disagreements over deliverables, timelines and costs. When the government finally pulled the plug on the project in 2011, the total cost had ballooned to £12bn, with little to show for it. The key takeaway here – poorly defined contracts can lead to disastrous legal battles. It's essential to have clear, detailed agreements that outline responsibilities, deliverables and consequences for failure. Next up is the Facebook-Cambridge Analytica data scandal, which highlighted the legal risks surrounding data privacy in IT projects. In 2018, it was revealed that the political consulting firm Cambridge Analytica had harvested personal data from millions of Facebook users without their consent. This scandal led to multiple lawsuits, regulatory investigations and a massive fine of $5bn imposed on Facebook by the Federal Trade Commission, FTC, for violating user privacy. The incident also triggered global scrutiny of data privacy laws and contributed to the strengthening of regulations like GDPR. This

case highlights the importance of adhering to data protection laws and ensuring that user data is handled ethically and legally. With today's stringent privacy regulations, the legal risks of non-compliance are simply too high to ignore. Our third case study takes us into the world of intellectual property with the Oracle vs Google legal battle. This lawsuit filed in 2010 centred around Google's use of Java APIs in its Android operating system. Oracle claimed that Google's use of these APIs without a licence amounts to copyright infringement. After a decade of legal battles, the case reached the US Supreme Court, which ultimately ruled in Google's favour in 2021. The court decided that Google's use of Java APIs constituted fair use under copyright law. The legal takeaway? Intellectual property protection is critical in IT projects, especially when dealing with software development. It's essential to ensure that all IP rights are properly negotiated and licensed to avoid costly disputes down the line. So how do you protect your IT projects from these legal risks? Here are some key legal protections that every IT project should have in place. Comprehensive contracts. Make sure your contracts clearly define project scope, deliverables, timelines and costs. Include provisions for change management so that any adjustments to the project can be handled smoothly. IP agreements. Clearly define who owns the intellectual property being created and how it can be used. This is especially important when working with external contractors or third parties. Data protection compliance. Ensure that your project complies with all relevant data protection laws, such as GDPR in Europe or CCPA in California. This includes having the necessary data processing agreements in place with third-party vendors. Liability clauses. Include clauses that limit liability in the events of delays, failures or breaches. This can protect your company from taking the full financial hit if something does go wrong. While legal protections are crucial, it's also important to follow best practices that can help you avoid legal pitfalls before they arise. Involve legal teams early. Make sure that your legal team is involved from the very start of the project. They can help to identify potential risks and draft the necessary legal agreements. Regular audits and compliance checks. Conduct regular audits to ensure that the project is adhering to all relevant laws and regulations, especially when dealing with data privacy and compliance. Document everything. Keep detailed records of all decisions, changes and communications throughout the project. This can be invaluable in resolving disputes later on. Train your teams. Make sure your project managers, developers and contractors are all aware of the legal risks and responsibilities involved in the project. Proper training can help prevent mistakes that lead to legal trouble. As technology evolves, new legal challenges are emerging. Artificial intelligence, blockchain and quantum computing are all areas where legal frameworks are still being developed. For example, in AI development, questions about algorithmic bias and liability for decisions made by autonomous systems are raising legal concerns. The legal landscape around these technologies is still constantly changing, which means companies need to be proactive about understanding and managing these emerging

risks. In conclusion, handling legal risks in IT projects is all about preparation and protection. By involving legal teams early, drafting comprehensive contracts, ensuring compliance with data protection laws and safeguarding intellectual property, you can avoid many of the common legal pitfalls that have derailed projects in the past. As IT projects become more complex, the legal risks will only grow. But by following best practices and staying aware of emerging challenges, you can navigate this legal minefield and deliver successful projects.

Welcome to this short session on inclusive computing, where we look at how to build technology that's accessible to everyone. From software design to hardware development, inclusive computing ensures that nobody is left behind in the tech revolution, no matter their ability, background, or identity. In the next few minutes, we'll look at what inclusive computing is all about, why it matters, and a few cool examples of how it's changing the world. So what is inclusive computing? At its core, it's about designing technology that can be used by the widest range of people, regardless of their abilities or circumstances. This means accounting for different physical, cognitive, and sensory abilities, as well as language barriers, socioeconomic factors, and cultural diversity. It's not just about making tech available to everyone, but ensuring it's usable and enjoyable for everyone. Why does this matter? Well, about 15% of the global population is with some form of disability. If our technology isn't inclusive, that's potentially 1 billion people who can't fully access or benefit from the tools that we're building. And it's not just about disabilities. Think about language differences, access to technology in remote areas, or even how tech is designed for different cultural contexts. Inclusivity leads to better innovation, because when you design for everyone, you push the boundaries of what's possible. Let's look at a couple of examples of inclusive computing in action. Microsoft's Xbox Adaptive Controller. Microsoft developed this controller for gamers with limited mobility. It features large buttons and ports to accommodate a wide range of input devices, proving that gaming should be accessible to all. Google's Project Euphonia. This project uses AI to help people with speech impairments communicate more effectively with voice recognition software by training the AI on non-standard speech patterns. These examples show how big tech is stepping up to make sure no one gets left out of the digital world. To create inclusive computing, we need to follow certain principles, like Flexibility in use. Tech should offer multiple ways to interact with it. Think of how websites offer both text-based and voice controls. Simple and intuitive. Make interfaces that are easy to understand, even for someone who's never used them before. Perceptible information. Provide information in multiple formats. For example, adding captions to videos helps not only people with hearing impairments, but also those in noisy environments. By embedding inclusivity from the start, we ensure that technology benefits the widest range of users. Inclusive computing isn't just a trend. It's the future of tech development. When we design for

everyone, we create better, more innovative, and more accessible technology. Whether it's giving gamers with disabilities a way to play, or creating voice recognition that understands everyone, inclusive computing makes sure no one is left behind in the digital age. Thanks for joining me on this quick tour of inclusive computing. Remember, inclusivity isn't just about ticking boxes. It's about building a future where technology works for all of us.

Welcome to today's lecture on the ethical impact of technology. It's no secret that technology has transformed nearly every aspect of our lives, but with great power comes great responsibility, ethically speaking. Whether it's AI deciding who gets a loan or social media shaping elections, technology creates some tough ethical dilemmas. Today we'll explore the most significant ethical impacts of modern technology, look at some case studies where things went very wrong, and discuss how we as technologists and users can navigate these issues. I've intentionally blurred the lines a bit between some of these topics because I want you to be introduced and reintroduced to these elements on a regular basis. We'll continue this process of exploring different concepts at both granular and abstract levels of depth in order to really see the bigger picture. Technology has both positive and negative effects. On one side, it has enabled incredible advances in medicine, communication, and education. But on the other side, it's responsible for privacy breaches, job automation anxiety, and widening the digital divide. Ethical questions aren't just about what's technologically possible, they're about what's right. Just because we can create AI systems that track every move you make online doesn't mean we should. Or should we? Let's start with privacy. In the digital age, privacy has become one of the biggest ethical concerns. Think about how much personal data is collected by companies like Google, Meta, or even your fitness apps. In fact, a study by Pew Research found that 79% of Americans are concerned about how their personal data is being used by companies. For many, it's an unknown about what data is being collected and how it's being utilised. But it's not just companies. Governments are in on it too. Edward Snowden's revelations about the National Security Agency's surveillance programmes show just how far governments can go in tracking citizens. This raises a critical ethical question. How much privacy are we willing to trade for security or convenience? I would argue that we've already conceded a great deal, but we're also seeing a pushback. One of the most infamous cases of privacy invasion is the Facebook-Cambridge Analytica scandal. In 2018, it was revealed that Cambridge Analytica had harvested data from millions of Facebook users without their consent to create targeted political ads during the 2016 US presidential election. The ethical breach here wasn't just about violating privacy. It was about manipulating public opinion. As one user famously put it in a post after the scandal broke, We're not Facebook's customers, we're its product. The scandal led to global outrage, fines and new regulations like the GDPR, the General Data Protection Regulation, in Europe. But

it raises a key ethical question. How do we balance technological innovation with user consent and transparency? Many systems are allowed to exist untethered, and that has risk all by itself. Next up is bias and discrimination in AI systems. AI is increasingly making decisions about important aspects of our lives. For example, who gets a loan, or who's hired for a job. But these systems are only as unbiased as the data they're trained on. And guess what? Our data often contains human biases. Take for instance Amazon's AI recruiting tool that was found to discriminate against women. The algorithm had been trained on resumes submitted over the past 10 years. Resumes which predominantly came from men. As a result, the AI penalised resumes that included the word women, or were from women's colleges. This demonstrates the fundamental flaws in many of these systems. Another example of bias in technology is the racial bias in facial recognition software. A study from MIT found that popular facial recognition systems, such as those used by law enforcement, had an error rate of around 34.7% when identifying dark-skinned women, compared to less than 1% for lighter-skinned men. These kinds of errors aren't just inconvenient, they're dangerous. In 2020, a black man named Robert Williams was wrongfully arrested in Detroit because facial recognition software mistakenly identified him as a suspect. This raises ethical concerns about discrimination and accountability in AI. How can we ensure AI systems are fair and transparent when they're increasingly being used in high-stakes situations? People put a huge amount of trust in these systems, but they very often get it badly wrong. Let's move to the economic side of tech ethics. Automation and job displacement. As AI and robotics become more advanced, more jobs are at risk of being automated. According to a report from Oxford University, up to 47% of jobs in the US could be automated in the next two decades. This isn't just about robots replacing factory workers, it's about automation affecting white-collar jobs, from accountants to radiologists. The ethical question here is, who benefits from automation? Are we building a future where only tech companies profit while millions of workers are displaced? Or are we finding ways to retrain workers for new, tech-enabled jobs? Many of you will work on automated solutions using libraries like Selenium. Would you be as willing to develop these tools if you knew they were making people jobless? Here's an interesting example from the world of automation. Self-driving cars. These vehicles are poised to reduce traffic accidents, but they also present a classic moral dilemma. Imagine a scenario where a self-driving car has to decide between hitting a pedestrian or swerving and putting the passenger's life at risk. This is known as the trolley problem, but in real life. Should the car prioritise the life of its passenger, or should it try to minimise overall harm? How should the algorithms be programmed to make these decisions, and who gets to decide the ethical framework? Companies like Tesla and Waymo are grappling with these very questions as they develop autonomous vehicles. Would you be more likely to buy a vehicle that prioritised your life over someone else's? Of course. What about if this is only a premium option to a handful of wealthy

individuals? Doesn't seem fair, does it? Finally, let's talk about the digital divide. While technology has the potential to democratise information and access, it's also creating a wider gap between those who have access to those technologies and those who don't. In a world where 60% of the global population has internet access, that still leaves billions without the ability to participate in the digital economy. Even within wealthy nations, there's a divide between urban areas with high-speed internet and rural communities struggling with poor connectivity. This raises questions about equity and access. Is it ethical to create a world driven by technology when large parts of the population are left behind? The COVID-19 pandemic exposed the digital divide like never before. As schools moved online, millions of students without reliable internet access or devices were left behind. In the US, an estimated 16 million students lacked the tools needed for remote learning. At Goldsmiths, we barely saw an impact in the computing department, but I saw many of my colleagues in other departments struggle to make the transition to online only. Both educators and students missed out, but at least they had the resources to mitigate many of these issues. This isn't just a technological issue, it's an ethical failure. We need to ensure that as we move towards a more digital world, education, healthcare and economic opportunities are accessible to everyone, not just the tech-savvy or well-connected. So how can we mitigate these ethical risks? Here are a few strategies. Ethical design principles. Companies need to implement ethical design frameworks that prioritise transparency, fairness and privacy from the start. Diversity in tech development. Building diverse teams of developers and decision makers can help reduce biases in the technologies that we create. Stronger regulation and accountability. Governments do need to step in with stronger regulations like the GDPR in Europe to ensure companies are held accountable for how they handle data and how they use AI. While other places in the world might be able to innovate more freely, they remain in a space where this uncertainty also poses risks. Public education. Today we need to educate the public on the ethical implications of technology so that they can make informed decisions about the products they use and the privacy they trade. You'll see some poor design practices if you study interaction design, but these are the sorts of things you need to be aware of if you are to understand the risk. In conclusion, the ethical impacts of technology are profound and complex. From privacy breaches to AI biases, from job automation to the digital divide, we must continually ask ourselves, how do we build a future where technology benefits everyone, not just the privileged few? The answers aren't always simple, but by putting ethics at the forefront of our innovations, we can create a world where technology uplifts society rather than divides it. This is a contentious topic and the discussion is an ongoing one, but it is important to at least try and address some of these complex issues.

Welcome to today's session on accessibility in software design. Today we're going to explore how accessible design makes technology usable for everyone, regardless of their abilities. Accessibility isn't just a technical feature, it's a human feature. The goal? To ensure no one is excluded from using the apps, websites and tools we build. In this lecture we'll look at a case study to see how one major company tackled accessibility challenges, and we'll explore some best practices that can help ensure inclusive software design. Let's start with the basics. What is accessibility in software design? At its core, accessibility means designing technology that can be used by people of all abilities, including those with physical, sensory, cognitive or learning disabilities. The World Health Organization reports that 1 billion people globally live with some form of disability. That's about 15% of the population. So, if your software isn't accessible, you could be excluding a huge number of potential users. But it's not just about numbers. It's about creating equal access to the digital world. As Tim Berners-Lee, the inventor of the web, famously said, the power of the web is in its universality. Access by everyone, regardless of disability, is an essential aspect. One of the best examples of a company embracing accessibility in software design is Microsoft. Over the last decade, Microsoft has placed accessibility at the core of its design philosophy, culminating in features that support users with disabilities across its products, whether it's Windows, Office or Xbox. In 2019, Microsoft introduced a feature in Microsoft Word and Outlook called Immersive Reader, designed to support people with dyslexia and other reading disabilities. Immersive Reader allows users to break down text into simpler chunks, change font sizes, spacing and even highlight syllables to improve readability. What's more, it includes a read aloud feature that uses AI to read the text in clear, natural speech. According to Microsoft, our mission is to empower every person and every organization on the planet to achieve more. That means everyone, including the one billion people with disabilities. After releasing Immersive Reader, a study by the British Dyslexia Association showed that students using the tool improved their reading comprehension significantly. The tool wasn't just useful for people with dyslexia, it turned out to benefit all users, proving that accessible design can help everyone. This concept is known as universal design, where products are designed to be accessible to the widest possible range of people. It's a win-win. As a designer, you not only make your product more inclusive, but you also make it better for all users. Obligations in accessibility. It's not just ethical to create accessible software, it's also the law. Many countries have strict regulations around digital accessibility. For instance, the Americans with Disabilities Act, ADA, requires that businesses and public services make their websites and digital products accessible to people with disabilities. In the European Union, the Web Accessibility Directive mandates that public sector websites and apps meet accessibility standards. Failure to comply can lead to hefty fines. In 2019, Domino's Pizza found this out the hard way. A blind man, Guillermo Robles, sued the company because he was unable to order pizza from their website and mobile app using a screen reader. The

U.S. Supreme Court sided with Robles, highlighting that companies must ensure their digital platforms are accessible. This ruling set a legal precedent and it's a major warning for businesses. Let's look at some key accessibility features that should be integrated into any software design. Screen reader support. For users with visual impairments, screen readers like JAWS or NVDA read aloud what's on the screen. Websites and apps need to be compatible with these tools, including providing alt text for images and ensuring text is properly structured. Keyboard navigation. Some users can't rely on a mouse due to motor impairments, so your software should be fully navigable with just a keyboard. Every interactive element, buttons, links, menus, should be reachable using the tab key. Color contrast and text size. People with low vision or color blindness benefit from high contrast text and the ability to adjust font sizes. Tools like WebAIM's contrast checker help ensure that text is legible for all users. Captions and transcripts. For people with hearing impairments, video content should include captions. According to a Facebook study, 80% of viewers are more likely to watch a video to completion when captions are available. Not just people with hearing impairments, but also users in noisy environments. Let's circle back to the Domino's case for a moment. The core issue wasn't just that the website was inaccessible, it was that Domino's didn't think digital accessibility was a priority. Their defense argued that the ADA written in 1990 didn't explicitly apply to the internet. But the courts disagreed and the lesson for all businesses is clear. Digital accessibility is no longer optional. Just like you make sure your physical store has a wheelchair ramp, your online platforms must be designed for accessibility. As a court case concluded, the failure to make digital platforms accessible violates the basic rights of individuals with disabilities under the ADA. So don't wait for a lawsuit to remind you. Start designing with accessibility in mind from day one. Let's not forget the benefits. Designing accessible software doesn't just protect you from legal risks, it also opens your product to a larger audience. Accessible design leads to higher user engagement. When more people can use your product, engagement naturally increases.

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Customer loyalty. Users appreciate companies that consider their needs and they're more likely to stick around if they feel included. Innovation. Accessibility features often lead to innovation that can improve the experience for everyone. The classic example? The dropped curb. Originally designed to help wheelchair users, it ended up being useful for parents with strollers, cyclists and delivery people. In the world of software, the same principles apply. Voice recognition and text-to-speech tools were originally designed for accessibility, but now they're used by millions for convenience. The future of software design is inclusive by default. As technology evolves, so do the tools we use to ensure accessibility. AI-powered features like real-time transcription are becoming more common, and platforms like YouTube and Zoom now offer automatic captioning to

ensure everyone can follow along. It's important that we continue to innovate, not just because it's the right thing to do, but because it's smart business. Accessible design isn't a niche concern. It's a crucial part of creating products that work for everyone.

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To wrap up, accessibility in software design is about creating inclusive, usable experiences for everyone, regardless of ability. Companies like Microsoft are leading the way with accessible features that benefit millions of users, while cases like Domino's remind us of the legal and ethical responsibilities that come with building digital products. Remember, accessibility isn't just about meeting requirements. It's about opening the door to a more equitable and innovative digital future. By designing with accessibility in mind, you create better software for everyone.

Welcome to today's session where we're diving deep into one of the most revolutionary pieces of gaming hardware, the Xbox Adaptive Controller. Launched by Microsoft in 2018, this controller was designed specifically for gamers with limited mobility, making video games more inclusive than ever. Today we'll explore how this innovative device is reshaping the gaming industry, how it works, and even discuss some of the potential ways it could be misused by players without accessibility needs. Let's get started. Before we get into the specifics of the Xbox Adaptive Controller, let's talk about why it was needed in the first place. Gaming has become a huge part of modern culture, but for years gamers with disabilities have been excluded due to limitations of traditional controllers. Complex button layouts, rapid response triggers, and precise movements make gaming nearly impossible for individuals with mobility challenges. In the US alone, the Entertainment Software Association reports that 46 million gamers have disabilities, yet up until recently the gaming industry hadn't made significant strides towards inclusivity. Enter the Xbox Adaptive Controller. Released in 2018, this device was designed by Microsoft in collaboration with organizations like the AbleGamers charity and SpecialEffect. The Adaptive Controller allows users with limited mobility to customize their gaming setup with external switches, buttons, mounts, and joysticks that can be plugged into the device. So what sets it apart? The Adaptive Controller is all about modularity and customization. Traditional controllers have a fixed design, an array of buttons, sticks, and triggers, but the Adaptive Controller lets users map controls to any external device that fits their physical needs. Let's break down some of the key features of the Xbox Adaptive Controller. Portability and modularity. The controller has 19 3.5mm jacks and USB ports, allowing users to connect a range of external devices like foot pedals, chin operator joysticks, and oversized buttons. This means players can create a completely personalized setup depending on their specific needs. Large buttons. The controller has two large programmable buttons on the face of the device.

These buttons are designed for gamers who struggle with smaller traditional gamepad buttons. They can be remapped to any function within the game. Compatibility with mounting. The device is designed to work with wheelchairs, stands, and other mounts, making it accessible to a wide range of gamers. Software customization. The Xbox Adaptive Controller works seamlessly with Xbox Accessories App, allowing users to remap controls, adjust sensitivity, and customize their setup to fit different games. The Xbox Adaptive Controller has transformed gaming for people with disabilities. It's not just a controller, it's a bridge to an entire world of entertainment and community. Gamers who previously struggled to engage with their favorite titles now have the tools they need to play at their own pace and comfort level. Now let's talk about the ethical implications of devices like the Xbox Adaptive Controller. Accessibility in gaming is not just about adding cool features, it's about recognizing the rights of individuals with disabilities to fully engage with the digital world. The Adaptive Controller aligns with the principles of universal design, which means creating products that are usable by everyone, regardless of their abilities. It shows that when companies invest in inclusivity, it's not just good for business, it's a step toward a more equitable society. Of course, as with any technology, there's always the potential for misuse. While the Xbox Adaptive Controller is designed to level the playing field for people with disabilities, some able-bodied players may attempt to use it for unfair advantages in gaming. For example, because the controller allows extensive remapping and customization, competitive players could potentially use it to streamline complex button combinations or rapid-fire actions, giving them an edge in games where precision and timing are key. Imagine a player using foot pedals for repeated actions like sprinting or shooting, actions that require more intricate inputs on a traditional controller. While this is an unintended use of the technology, it highlights the ongoing ethical debates. When does customization cross the line into unfair play? Developers and gaming platforms need to monitor these potential abuses, while ensuring that the device remains open and flexible for its intended users. So how can the gaming community prevent this kind of misuse? A few approaches include. Fair play policies. Gaming companies could introduce rules that restrict how players can use the Xbox Adaptive Controller in competitive environments, while allowing full use for those with accessibility needs. Monitoring competitive play. Esports leagues and competitive platforms might need to implement software tools that flag unconventional controller setups or input patterns, similar to how anti-cheat software currently works. Educating the gaming community. Education is key, ensuring players understand that the Adaptive Controller is designed to make gaming accessible, not to give able-bodied players a competitive edge, can foster respect for the tool's true purpose. The success of the Xbox Adaptive Controller has spurred other companies to think more about accessibility in gaming. For example, Sony has worked to make its PlayStation consoles more accessible, introducing features like custom button mapping and modularity. Logitech launched a line of adaptive gaming accessories that work with

the Xbox Adaptive Controller, further expanding customization options for players. The impact of Microsoft's innovation has rippled across the industry, pushing the gaming world toward more inclusive practices. As we look to the future, the Xbox Adaptive Controller is just the beginning. AI-powered tools that adapt to a player's abilities in real-time, VR controllers that provide haptic feedback for various impairments, and even gesture-based controllers that remove the need for traditional input methods are all on the horizon. The real success of the Xbox Adaptive Controller isn't just in what it does today, but how it's inspiring the next generation of inclusive tech for gamers around the world.

Welcome to today's lecture on the BBC's accessibility strategy, a pioneer in making digital content more accessible to everyone. As one of the most prominent broadcasters in the world, the BBC has long been at the forefront of developing technologies and strategies that ensure inclusive access to media for all viewers. I had the privilege of working closely with the BBC during my time as a research fellow at the Nottingham University's Mixed Reality Lab, where I got to see some of these exciting accessibility innovations firsthand. Today we'll explore how the BBC has tackled the challenge of creating accessible content, the cutting edge technology they've developed, and how it's setting a benchmark for media companies globally. Let's start by addressing the big question. Why is accessibility so important in broadcasting? The BBC is a public service broadcaster, and part of its mission is to ensure that everyone, regardless of physical or cognitive ability, can access their content. With over 11 million people in the UK living with some form of disability, according to the UK's government's Office for National Statistics, the BBC recognised early on that it needed to cater to a diverse audience. And it's not just about meeting legal requirements, it's about fostering an inclusive society where everyone can participate in and enjoy the same cultural experiences. The BBC's accessibility strategy covers a range of areas, from television programming to online content, and even interactive experiences. Let's break down some of those key components. Subtitles and Closed Captioning. The BBC provides subtitles for nearly all of its TV programmes, including live broadcasts. They've pioneered AI-based real-time captioning to make this possible. Audio Description. For people with visual impairments, the BBC offers audio descriptions, where a narrator describes important visual elements of a show. This service is available on many BBC programmes, including hit shows like Doctor Who and Peaky Blinders. Sign Language. The BBC broadcasts a significant number of shows with British Sign Language interpretation, especially for news and children's programming. Accessible iPlayer. The BBC's digital platform, BBC iPlayer, has been designed with accessibility in mind, offering keyboard navigation, voiceover support and customisable subtitles. Users can adjust text sizes, contrast and other features to suit their needs. One of the most exciting projects I got to witness during my time working with the BBC was the development of the accessible BBC iPlayer. The

team I worked with at Nottingham's Mixed Reality Lab was focused on pushing the boundaries of how users interact with digital content, and the BBC was a perfect partner for exploring new ways to make online media more inclusive. BBC iPlayer has been a game-changer for accessibility in the digital space. Not only does it offer a fully customisable experience for users with disabilities, but the platform also integrates features like screen reader compatibility and keyboard shortcuts. The platform is continually evolving, with updates that improve accessibility based upon user feedback. The BBC iPlayer team collaborated with GDS, Government Digital Services, to meet and exceed WCAG 2.1 standards, ensuring that the platform is accessible to all. Subtitling has always been a cornerstone of accessibility, but the BBC has taken it a step further with real-time captioning technology. For live broadcasts, like sporting events or political debates, accurate and fast subtitling can be a challenge. The BBC uses AI-driven speech recognition to generate real-time subtitles, reducing the delay between spoken words and captions appearing on screen. This AI-driven approach to subtitling has been groundbreaking. A study by Ofcom revealed that BBC live subtitling had an accuracy rate of over 98%, far surpassing industry standards. This is crucial, particularly in news and current affairs, where access to real-time information is critical. For people with visual impairments, audio descriptions are essential for enjoying TV shows and films. The BBC has committed to providing audio descriptions for at least 20% of its content, but many popular programmes far exceed this benchmark. I had the chance to sit with a focus group of visually impaired viewers during my collaboration with the BBC, where we evaluated the effectiveness of audio descriptions on some of their drama series. One participant noted, Audio description gives me the freedom to enjoy my favourite shows independently without having to ask someone to explain what's happening. This speaks to the real-world impact of accessible design, empowering users to have full autonomy over their media consumption. The BBC has always been forward-thinking in accessibility and their research and development department is constantly working on new innovations. One project that particularly stood out to me was their experimentation with haptic feedback and wearable technology to make TV more immersive for people with sensory impairments. Imagine watching a football match and being able to feel the rumble of a crowd's roar through your wristband. That's the kind of innovation the BBC is exploring. They've also been investigating how voice-controlled interfaces can make accessing content easier for people with mobility challenges, ensuring that platforms like iPlayer remain at the cutting edge of accessibility. Beyond the technological advancements, the BBC has a strong ethical and legal responsibility to provide accessible content. The UK Equality Act 2010 requires public institutions to make reasonable adjustments to ensure that disabled people can access services. The BBC has not only met these requirements but often exceeds them, setting the gold standard for other broadcasters to follow. However, accessibility isn't just about ticking legal boxes, as the BBC themselves say. We believe

in removing barriers and it's our duty as a public service broadcaster to ensure our services are available to everyone. As the media landscape evolves, so too will the tools and technologies used to make content more accessible. In the future, we're likely to see even more personalisation, where users can adjust not just the subtitles and audio descriptions, but interface layouts, content formats and even the pace at which information is delivered. One promising area is immersive technologies like virtual reality and augmented reality, where the BBC is already experimenting with accessibility features. For example, augmented reality could be used to overlay real-time sign language interpreters or subtitles onto live events, making them more accessible in dynamic ways. Of course, accessibility is an ongoing journey. The BBC faces challenges like ensuring its platforms remain accessible across a wide range of devices, from smartphones to smart TVs. As technology evolves, new accessibility issues will emerge, but the BBC's commitment to inclusive design ensures that they're always at the forefront of innovation. Additionally, continuing to engage with disabled users in testing and feedback loops will be key to refining accessibility features over time. In conclusion, whether it's through real-time captions, audio descriptions or developing the next generation of accessible digital platforms, the BBC continues to innovate in ways that empower users of all abilities.

Welcome to today's video on implementing inclusive practice for computer scientists. In tech, inclusivity isn't just a buzzword, it's a key part of building ethical, sustainable and innovative technology. Whether you're designing software, working on AI or developing the next big app, inclusivity ensures that everyone benefits from what you're building. Today we'll cover some practical strategies for implementing inclusive practices in your work. We'll look at tools, methodologies and best practices that you can start using today with examples from industry leaders like Microsoft, Google and others who are doing inclusivity right. Before we look into the how, let's quickly talk about why inclusivity is so critical in technology. According to McKinsey, companies with higher diversity are 35% more likely to have financial returns above their industry's median. It's not just about representation, it's about better decision making, broader market reach and more innovative solutions. As Tim Cook, CEO of Apple once said, the most diverse group will produce the best product. If you're going to have a creative environment like Apple's, you need different perspectives. In computer science, inclusivity ensures that we're designing technologies that don't exclude people based on gender, race, disability or any other factors. Goldsmiths holds the same ethos. We have people from all walks of life coming together to do amazing things. Let's jump into our first strategy, inclusive design thinking. This approach involves thinking about inclusivity from the very beginning of the design process. It means considering a wide range of user needs and experiences, including those of people with disabilities, cultural differences and various cognitive abilities. Take Microsoft's inclusive design toolkit, which encourages

developers to ask questions like who is excluded by this design and how can we extend this to include more people? By building inclusivity into the design process, you end up with solutions that work for everyone. One of the key principles here is design for one, extend to many. This means that by designing for users who might face the greatest barriers, such as people with disabilities, you can create solutions that improve the experience for a wider audience. Next, let's talk about the importance of diverse development teams. The more diverse your team is, the more perspectives you bring into the development process. This diversity helps you identify any problem areas early on and think creatively about how your technology can serve all users. For example, Google uses diverse hiring practices to ensure its teams reflect the global user base they serve. By bringing in people from different backgrounds, experiences and abilities, they are able to design products like Google Maps that consider accessibility features such as wheelchair-friendly routes and voice navigation for people with visual impairments. Quick tip, when forming teams, actively seek out underrepresented groups in tech. Whether it's women, people of colour or people with disabilities, more perspectives means more inclusive solutions. A great practical approach to inclusivity is putting accessibility first in your software development. This involves not treating accessibility as an afterthought or something to add in later, but adding it into the core functionality of your product from the start. One best practice here comes from the Web Content Accessibility Guidelines, or WCAG, which provide standards for ensuring that websites are accessible to all users. Things like high contrast colour schemes, keyboard navigability and screen reader compatibility should be considered from the outset. Consider Slack, which has introduced features like these. These features not only benefit people with disabilities, but also improve the overall user experience for everyone. One of the most pressing challenges in tech right now is algorithmic bias in AI systems. AI models are only as good as the data that they're trained on. And if that data contains biases, the AI will reflect and possibly even amplify them. To combat this, computer scientists need to implement bias detection and mitigation techniques during the development of AI systems. For example, IBM has developed the AI Fairness 360, a toolkit designed to help developers detect and mitigate bias in machine learning models. This tool allows developers to test their algorithms against fairness metrics, ensuring that the technology doesn't unfairly disadvantage certain groups of people. To quote Joy Buolamwini, a computer scientist and founder of the Algorithmic Justice League, AI is not neutral. We have a responsibility to ensure it reflects the best of humanity, not the worst. Inclusive development doesn't stop at design and coding. It's equally important to conduct user testing with a diverse group of people. This ensures your product works for a wide variety of users, including those with disabilities, different languages, and various levels of technical literacy. Consider the way that Apple tests its accessibility features like voiceover and magnifier with a variety of users, including people with visual impairments. By testing in real-world conditions and gathering feedback from diverse

users, Apple ensures that their devices are usable by all. The key here is to listen. Your users are your greatest resource when it comes to spotting issues and opportunities for inclusivity. Another practical strategy is writing inclusive code. This means ensuring that the code you write can be easily adapted to support different languages, screen sizes, and accessibility tools. Things like properly labelled HTML tags for screen readers, or using variable names that are clear and neutral contribute to making your software more inclusive. For example, when working on websites, ensuring that all images have descriptive alt text allows screen readers to describe the content to users who are visually impaired. Similarly, using ARIA roles, accessible rich internet application roles, in your web design helps people who rely on assistive technology navigate your application more easily. The best part? Writing inclusive code isn't difficult. It's just about being mindful. Accessibility isn't a set-it-and-forget-it approach. It's something that must be continuously improved. Offering ongoing training programs for developers is essential in staying updated on the best practices for accessibility and inclusivity. Many companies, including Adobe, offer internal training programs where developers learn how to incorporate inclusivity into their daily work. This might include training on unconscious bias, learning how to test for accessibility, or understanding the diverse needs of users. To sum up, inclusivity isn't just a box to tick. It's a responsibility and an opportunity. By following these strategies, using inclusive design thinking, forming diverse teams, prioritizing accessibility, mitigating bias in AI, testing with diverse users, writing inclusive code, and continuously learning, you can create technology that serves everyone. Thanks for joining me today. Now it's time to put these strategies into practice and start building a more inclusive future in computer science.

Welcome to today's lecture on assessing ethical impacts in technology. In a world where tech is evolving faster than ever, ethical considerations have become critical to ensuring we're not only building powerful tools, but also responsible ones. Whether it's social media platforms, shaping public discourse, AI systems making decisions for us, or data being collected and sold behind the scenes, we need to ask, what are the ethical impacts of the technology that we create? Today we're going to focus on how we can assess these impacts, including holding companies accountable for their actions. We'll dive into real world examples like social media platforms, and discuss best practices for embedding ethics into tech development. Let's get started. So why do we need to assess the ethical impacts of technology? Technology is integrated into every part of our lives, from the apps we use, to the AI that recommends content, to the algorithms that decide who gets a loan. As creators of these technologies, computer scientists have a responsibility to think about how their work affects society. A study by Pew Research found that 79% of Americans believe that tech companies need to be held accountable for the products that they create. Think about it. Algorithms can reinforce biases, social media can spread misinformation, and data breaches can expose personal information.

Ethical assessments help us mitigate these risks. They force us to ask tough questions like, is this technology being used for harm? And who benefits, and who might be excluded? With great power comes great responsibility, and nowhere is that more true than in tech. Let's start by looking at one of the most visible ethical issues in tech today. Social media accountability. Platforms like Facebook, Twitter, and YouTube have reshaped the way we communicate, but they've also been criticised for allowing the spread of misinformation, hate speech, and polarising content. In the Facebook Cambridge Analytica scandal, where millions of users' data was harvested without consent to influence elections, this wasn't just a breach of privacy, it raised ethical questions about the role of social media in manipulating public opinion. Mark Zuckerberg has testified multiple times before Congress, promising reform, but the ethical questions remain. How do we hold these platforms accountable? Social media companies are now facing pressure to fact-check content, promote transparency, and curb harmful behaviours. But are these steps enough? Holding platforms accountable is no easy task. It's a balancing act between free speech and preventing harm. Here are some strategies tech companies and regulators can use to assess the ethical impacts of social media. Content moderation. Platforms can implement stronger moderation systems that use both AI and human reviewers to flag inappropriate content. For example, YouTube has been improving its AI system to better detect hate speech and misinformation, but it still relies on human oversight to make the final decisions. Transparency reports. Social media platforms should regularly publish transparency reports that detail how they can handle user data, what kind of content is being removed, and why. This increases accountability and gives users insight into the platform's decision-making processes. Independent audits. Independent third-party audits can be conducted to assess whether platforms are complying with ethical standards. This includes evaluating algorithms for bias and examining how user data is used and stored. One practical tool for assessing the ethical impacts of technology is an ethical impact assessment. Similar to environmental impact assessments, EIAs provide a structured way to evaluate the ethical implications of a tech project before it is deployed. An EIA considers factors like privacy and data security, how is user data being collected, stored, and shared, what are the risks of a data breach, bias and fairness, is the technology designed in a way that avoids reinforcing bias or discrimination. Social impact. How will the technology affect different social groups? Will it create a divide between those who have access to it and those who don't? For example, AI development companies like OpenAI have begun integrating EIAs into their workflow to assess how their models could be misused or cause harm. Universities such as this one have ethics committees with independent assessment of risk for projects. Not just an afterthought, but something that could affect funding decisions for public research. Even companies with the best intentions can stumble when it comes to ethics. Take Google for example. In 2019, they established an AI Ethics Council to help

guide the development of their AI products. The council was tasked with assessing the social and ethical impacts of AI systems. However, the council quickly became controversial when it was revealed that some of the appointed members had conflicting interests and had made controversial statements about issues like climate change and diversity. Amid backlash, Google dissolved the council just a week later. This highlights an important lesson. Ethical oversight must be transparent, diverse, and genuinely committed to serving the public good. Simply creating a committee doesn't solve ethical problems. There has to be a clear plan for how ethics are integrated into the development process. Bias in AI systems is another area where ethical assessments are crucial. AI is used in everything from facial recognition to job recruitment, but biased data can lead to discriminatory outcomes. You may remember from an earlier lecture that Amazon's AI hiring tool was scrapped after it was discovered that it was biased against women. How many times have you failed a capture that was impossible or been unfairly challenged by technology? For some people, that's an everyday occurrence based on something as arbitrary as the colour of their skin. One way to assess and mitigate bias is by using bias detection tools like AI Fairness 360 from IBM, which I mentioned earlier. These tools evaluate AI models to ensure that they are not unfairly discriminating against certain groups. But beyond tools, diverse teams of engineers and researchers can help identify and reduce bias in AI systems. Joy Buolamwini, a leading voice in AI ethics, put it well, Algorithms are not neutral. We must actively work to make them equitable or they will reinforce the biases of society. We spend lots of time at Goldsmiths on decolonizing the curriculum, but so much of colonialism is deeply ingrained in contemporary pedagogical practices. It is really difficult to see things like a lack of diversity, representation, and sometimes even intrinsic bias. Ethical technology also requires strong regulations. Without oversight, companies can easily prioritize profits over people. That's why frameworks like GDPR, the General Data Protection Regulation in Europe, have been so impactful. GDPR requires companies to get explicit consent before collecting data, and it gives users the rights to access and delete their data. In the US, there's growing interest in passing similar legislation, especially after the fallout from Cambridge Analytica and the increasing power of AI systems. Laws like the California Consumer Privacy Act, CCPA, are steps in the right direction, but much more is needed to ensure tech companies are legally accountable for the ethical impacts of their products. So how can computer scientists ensure their work is ethically sound? Here are a few best practices for ethical tech development. Build diverse teams. As we've discussed, diversity in development teams helps reduce bias and ensures a wider range of perspectives. Conduct ethical impact assessments. Before launching a product, consider conducting an EIA to assess potential ethical risks, from privacy concerns to social inequality. Incorporate transparency. Whether through transparency reports or open source code, making your processes visible to the public is a critical way to build trust and demonstrate accountability. Engage with ethical frameworks.

Familiarise yourself with ethical frameworks like the IEEE's Ethics in Action or Google's AI Principles. These guidelines help developers make decisions that prioritise ethical considerations alongside technical goals. Sometimes looking at guidelines like those provided by the American Psychology Association can help us to understand contemporary challenges in areas of social computing and studies around usability. We must always be thinking critically about how we can do better. In conclusion, assessing the ethical impacts of technology isn't just an optional extra. It's a fundamental part of creating responsible, equitable systems, whether it's holding social media platforms accountable, mitigating AI bias or implementing ethical frameworks, we have a duty to ensure our innovations are a force for good. As Sheryl Sandberg, COO of Meta once said, we need to work together to ensure that our technology works for everyone, not just the privileged few. This was a hard topic to cover because many of my friends and colleagues have been victims to poor technological and design choices. It's not nice to think about people suffering. I hope that you will all think about these things throughout your career and try it in your own ways to make the world a better place.

How should companies incorporate diverse employee perspectives to ensure that the design and the implementation is inclusive? So I think the number one thing that employees and companies and organizations should do is focus on training. So they should provide cultural training to internal stakeholders, to teams that are specifically tailored for the company, the industry, or the products that they are trying to build and compute. So the number one thing is training. The number two thing is making sure that everyone is developed, different types of groups that have to do with inclusivity or that represent various aspects of society are involved as stakeholders in the decision-making process, either by co-designing parts of the application or by monitoring parts of the application. So the second thing is engaging employers and engaging stakeholders in order to making sure that everyone's voice is being heard and taken into accountability within this process. And the third thing that I think is extremely important is having this cultural and societal audit sort of way. That is extremely important to regularly review the systems and making sure that the usability of the systems and their computing applications matters across different types of groups. So I think that given that these have taken place, we can ensure that no bias, no discrimination, or no compliance issues will have been taking place in these applications. Are there any ethical or legal risks that you think are particularly associating with failing to account for diverse user experiences and workplace technology? Where there are specific frameworks and specific governance frameworks. In the UK, I think you have the Equality Act. In the UK, the EU has voted for the AI Act. So there are specific frameworks that can really create legislative damage sometimes if there is a huge problem or reputational damage in an organization. But I think that most importantly, what matters here is internal appreciation of products, but also catering to

users and vendors. So there are direct, there can be direct consequences and indirect consequences when organizations are not taking into account inclusivity in specific companies, specific products or services. How do you think that employers should ensure that their computing platforms and work technologies are accessible to neurodiverse employees?

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Again, I think the most important thing is human resources to really train on neurodiversity. It's something that in Greece, for instance, is not widely accepted or widely adopted by human resources departments in smaller companies. Larger companies have started training seminars. So the first thing is to train on neurodiversity and educate employers and educate stakeholders,

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line managers, extremely important about neurodiversity. The second thing that I would say is to invest in assistive technology or invest in platforms that can really cater towards different people's neurodiversity. Modern technology has the capability to really assist different types of people with different types of needs or approaches or personalized preferences in job or education with technology like text to speech, speech to text, and all these language assist capabilities that we have right now. So these organizations, I think it's really important to also invest into customizable solutions, whether they can build it in-house or whether they can partner with vendor companies in order to prioritize this accessibility that we need to have for our people in the workforce. And I think that this is extremely important because it can really help ensure equity and not only quality. And that's extremely, extremely important because as soon as we start focusing on equity, we will really see a change in inclusion and diversity. And I think that this is the direction to go from now on. In workplaces where computing systems are used to manage and monitor employee performance, how can employers ensure that these technologies do not exacerbate existing power imbalances, particularly for employees from marginalized groups? Well, that's a great question. Let's not forget that when we automate parts of our job and when we automate, for instance, parts of the hiring process in the earliest days of technology, we also saw from huge corporations falling a victim of bias, of algorithmic bias. So it's really important to go back and reevaluate. So the number one thing I think that's important is transparency. So having a one-to-one relationship with the employees, explaining the metrics used, explaining how algorithms are used, algorithmic transparency is extremely important as well, making sure that communicating the process and understanding and gaining input and feedback from employees whenever something may not be tailored or applicable to them. The second thing I think is important is to do regular audits on bias. So including the human-in-the-loop approach, anytime you automate something, it's extremely

important to audit it and monitor it and evaluate it. So with regular audits, it is important to ensure that nothing is left hidden and no discriminatory pattern emerges in a certain way. The third thing I think that we now have the capability to do is use technology as an audit tool. So use technology to direct it and generative AI gives us this capability to direct it to locate biases in our existing system management, in our existing hiring process, in our existing performance management. So we can use technology to help us locate the biases we have in our chain of pipeline in organizations. So I think that would be one of the actual positive directions that we can move together with technology.

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Companies can design computing systems and digital workplaces that account for the intersectionality of employees' identities, race, gender, disability, and what further practical steps can employers take to ensure that their technologies are inclusive at multiple levels and what risks do they face if they fail to do so?

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Well, the first thing, again, is the part of analysis and using these and using the intersectionality by design approach as we're using the responsibility by design approach or security by design approach in computing applications in modern AI. So the first thing is evaluate and analyze how identities that include subcategories like race, sex, gender, disability, yada, yada, yada, are included and are being catered in the process. The second thing is making sure that there is an inclusive design and an inclusive design means two things, principles and best practices and universal practices, but also an inclusive and diverse team that can really cater for these practices, that can really implement or can flag when one of these things is not taken into consideration. And the third thing is feedback loops and phase releases. So by releasing something in a phase rollout manner, we are able to control multiple steps. We're able to monitor. We can go back and correct something and then re-release it to different groups and different populations in order to locate any intrusive biases or any issues with intersectionality. So pilot testing a lot, regular updating a lot will make sure that there are no legal issues, no reputational damages to the organization, that every employee is satisfied, that there is no talent attrition whenever we're talking about internal stakeholders, that there is no end user attrition, that basically we built these products and these services with everyone in mind and taking into account different diverse groups that technology is now experiencing and it's talking to. To what extent should employers take responsibility for improving digital literacy among employees from diverse backgrounds? Can the lack of digital fluency, particularly among underrepresented groups, be considered an ethical issue for employers and what further measures should they implement to foster inclusion in the digital age? I think that

this question has a very, very interesting direction. I really agree with the undertones of this question. Digital literacy is crucial. Upskilling is crucial nowadays. It's becoming as important as basic standard literacy is becoming for employees. So an organization that needs their employees to be able to perform well and stay ahead of the latest trends is obligated and should invest in literacy and upskilling their employees. And I think that what's important here is to create tailored upskilling programs that invest in transformation. Also in cases of underrepresented groups, I think that there's a huge possibility for the organization to also increase their own ethical mark and the socials mark in the area but by communicating and collaborating with NGOs or with community organizations, give accessibility to skills, develop programs that can cater for different types of people in order to increase diversity. And increasing diversity, we see that organizations are really profitable. It's a huge opportunity for them to be making more profit when they increase diversity. So by incentivizing and by using digital skills and diverse formats, diverse languages, more accessible training to people that within the organization otherwise wouldn't have, it's important to bridge this literacy gap that we're seeing sometimes, but also making sure that no systemic inequities or inequalities that we're seeing will continue to take place. It's a profit for the individual. It's a profit for the employer, but it's also a profit for the organization and the end users or partners of that organization.

Welcome to today's video on Sustainability in Computing. Inspired by Gay Gordon Burns' TED talk, You Deserve the Right to Repair Your Stuff, we're going to explore the major sustainability challenges facing the tech industry. Small electronics like smartphones, laptops and tablets are in every part of our lives, but their environmental footprint is anything but small. We'll discuss the materials used in electronics, the environmental impact of production and what can be done to make computing more sustainable. Let's get into it. Most people don't think about the environmental cost behind their smartphones or laptops, but creating these devices involves extracting finite resources, producing toxic waste and generating significant amounts of CO2. According to the Global E-Waste Monitor, the world generated 53.6 million metric tons of e-waste in 2019, and that number is climbing. That's like throwing out 4,500 Eiffel Towers worth of electronics each year. What's more, only 17% of that e-waste is properly recycled. The rest? It ends up in landfills, often in developing countries, where it can leach harmful chemicals into the environment.

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One of the biggest sustainability challenges in computing is the use of rare earth elements. These materials like neodymium, tantalum and cobalt are critical for manufacturing small electronics. They're used in everything from the magnets in your smartphone speakers to the batteries that power your laptop. The problem? Extracting

these materials is highly resource-intensive and often leads to severe environmental damage. For example, mining rare earth elements in countries like China and the Democratic Republic of Congo has caused significant pollution and soil degradation. Mining for cobalt, used in lithium-ion batteries, also has a devastating human cost, with reports of child labour being a major issue in extraction sites. Let's talk about e-waste, one of the biggest sustainability challenges in tech. As consumers, we're upgrading our gadgets faster than ever. The average smartphone lifespan is just two years, and many laptops don't last much longer. When the devices reach the end of their life, they often end up in landfills, where toxic components like lead, mercury and cadmium can leak into the soil and groundwater. A report from the United Nations University stated that in 2019, e-waste generated $57 billion worth of recoverable precious metals, including gold and silver, but the vast majority of that material was lost because the devices weren't recycled properly. One of the biggest drivers of e-waste is planned obsolescence, a strategy used by manufacturers to design products with a limited lifespan, encouraging frequent upgrades. Whether it's slowing down old smartphones with software updates, or making devices with non-replaceable batteries, planned obsolescence is a huge sustainability problem. Gay Gordon Byrne emphasises this issue in her TED Talk. You deserve the right to repair your stuff, pointing out that companies deliberately make it difficult to repair or upgrade electronics. This forces consumers to buy new devices rather than extending the life of the ones they already have. This might be the reason I still use a desktop and Linux. I can upgrade incrementally without spending lots of money or throwing things away that might be useful. I also don't think my department head would be happy if I tried to buy a new machine every six months. The production of electronics is incredibly resource heavy. For example, creating a single smartphone uses 62 different metals, including precious metals like gold and rare earth elements. The manufacturing process requires large amounts of water, energy and chemicals. A 2016 report by the Green Electronics Council found that manufacturing a typical smartphone generates 85 kilograms of CO2. That's more than 10 times the weight of the device itself. The energy required to mine materials, transport components and assemble the product contributes significantly to climate change. We all want the newest gadgets, but at what cost? Fortunately, many tech companies are starting to address these sustainability challenges. Apple has committed to using 100% recycled aluminium in the construction of some of its products like the MacBook Air. They've also introduced recycling robots like Daisy, which disassemble old iPhones to recover valuable materials. Google has pledged to make all of its hardware products carbon neutral by 2030, and Microsoft is investing heavily in circular computing, aiming to be carbon negative by 2030. But while these initiatives are important steps forward, they only address part of the problem. Consumer behaviour and government policy also play a critical role. One important movement pushing back against planned obsolescence is the right to repair movement. Advocates argue that

consumers should have the right to repair their own devices or have them repaired by third parties without voiding warranties or facing major barriers from manufacturers. In recent years, legislation in the European Union and states like California have aimed to give consumers more power to repair their electronics. Gay Gordon Brown is a leading voice in this movement, advocating for laws that make it easier to fix broken devices rather than forcing people to throw them away and buy new ones.

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One way to combat the environmental impact of electronics is through the use of sustainable materials and by creating a circular economy for electronics. Companies are starting to look at biodegradable plastics, recycled metals and even plant-based materials to reduce the environmental cost of production. For example, Fairphone is a company that builds modular smartphones designed to be easily repaired using responsibly sourced materials and recycled components. Their goal is to extend the life of devices and reduce the need for constant upgrades. I own an acoustic guitar made from sustainable wood so that I know that I'm not contributing to the problem. These sacrifices are often worth it to motivate consumer markets to accommodate customers. Consumers also have a huge role to play in making computing more sustainable. Recycling old electronics responsibly is one of the most impactful ways to reduce e-waste. Many tech companies are now offering take-back schemes where consumers can return old devices for recycling. But it's not just about recycling. Extending the life of your electronics through repair and upgrading rather than buying new devices can help reduce demand for resource-intensive manufacturing. How many of you have fixed some tech that belongs to a friend, a colleague or a family member? Feels good, doesn't it? I believe that we can all contribute when it comes to making society better.

Welcome to this short video on energy efficient data centers, the backbone of our digital world. Whether it's streaming your favorite show, storing your files in the cloud, or running complex AI models, data centers make it all possible. But powering these digital giants comes with a significant energy cost. Today we'll look at how modern data centers are becoming more energy efficient and sustainable. We'll explore how they work, what makes them energy efficient, and some cool real-world examples like a bank that used its data center's excess heat to warm a swimming pool. Let's begin. Firstly, what exactly is a data center? A data center is a facility that houses computer systems, servers, storage devices, and networking equipment. These centers are essential for hosting websites, storing data, and running applications 24 hours a day, 7 days a week. As the demand for cloud computing grows, so does the need for data centers, but all that power doesn't come for free. It requires a lot of energy to keep the servers running and prevent them from overheating. Data centers are energy-hungry beasts, with

estimates suggesting they consume about 1% of the world's electricity. With the rise of big data, AI, and cloud services, this number is expected to increase. But it's not just the electricity that powers the servers, it's also the cooling systems that keep them from overheating. In fact, cooling can account for up to 40% of a data center's total energy usage. This is where energy-efficient data centers come in. These facilities use innovative designs and technologies to reduce the amount of energy they consume, making them more sustainable and cost-effective. So how do energy-efficient data centers actually work? It's all about optimization, making sure every watt of energy is used as efficiently as possible. Here are some key strategies that they use. Efficient cooling. Instead of traditional air conditioning, many data centers use liquid cooling systems, or free cooling, which takes advantage of naturally cool air or water from the environment to reduce energy use. Server virtualization. This technique allows multiple virtual machines to run on a single physical server, reducing the total number of servers needed and thus lowering energy consumption. Renewable energy. Many data centers are now powered by renewable energy sources, like solar and wind, reducing their carbon footprint. Waste heat recovery. Some data centers capture the heat generated by their servers and reuse it in creative ways like heating nearby buildings or swimming pools. Speaking of which, let's talk about a fascinating example. One of the coolest examples of energy-efficient data centers comes from innovations used at NatWest Bank in Exmouth, Devon, where they use the excess heat from a data center to heat a swimming pool. Here's how it works. The data center produces a lot of heat as it runs its servers, and instead of letting that heat go to waste, they install a system to capture the waste heat and transfer it to the local community swimming pool. This not only reduced the energy needed to heat the pool, but it also helped cool the data center more efficiently, effectively taking heat where it's unwanted and moving it somewhere where it is desirable. Just brilliant. This is a perfect example of how companies can get creative with energy efficiency, finding ways to reduce waste and give back to the community. NatWest Group isn't the only one leading the charge. Here are a few more examples of energy-efficient data centers around the world. Google's data centers. Google is one of the biggest players in this space. They've invested heavily in machine learning algorithms to optimize cooling and energy use, reducing energy consumption by 40% in some of their data centers. Google is also committed to running its data centers on 100% renewable energy. Cutting energy usage at source and then contemplating sustainable alternative sources is a double-pronged approach that provides real benefits. Microsoft's underwater data centers. Microsoft has been experimenting with underwater data centers like the Project Natick initiative, which uses a naturally cool temperatures of the ocean to keep servers cool. Not only does this reduce the need for traditional cooling systems, but it also increases energy efficiency by using renewable energy sources like offshore wind farms. These aren't active anymore, but it's a great proof of concept to show that original thinking can offer viable and possibly even better

alternatives. Facebook's Lul data center, located in Sweden. This facility takes advantage of the region's naturally cold climate to keep servers cool, using free air cooling systems. It's also powered by 100% renewable energy, primarily from hydroelectric sources. Makes sense, considering where they're situated. The future of data centers is about becoming even smarter and more sustainable. With increasing reliance on cloud services and AI, the need for energy efficient solutions will only grow. We're likely to see more innovations like AI driven optimization, waste heat reuse and alternative cooling methods. There's also a growing push for data centers to be powered entirely by green energy, reducing their carbon footprint even further. Incorporating energy efficient practices isn't just good for the environment, it's good for business. By reducing energy costs, companies can save money while also meeting sustainability goals. In conclusion, energy efficient data centers are a critical part of building a more sustainable digital future. From using liquid cooling and renewable energy to reusing waste heat, companies are finding innovative ways to reduce their environmental impact. With examples like NatWest Group heating a swimming pool and Microsoft experimenting with underwater data centers, it's clear that the future of computing can be both powerful and sustainable. Thanks for watching and remember, every click we make has an energy cost, so let's continue pushing for greener tech solutions.